Multiscale Sample Transformer for Hyperspectral Image Classification

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***Abstract*—Hyperspectral image (HSI) captures hundreds of narrow bands, offering abundant spatial and spectral information across various application domains. In recent years, deep learning techniques have shown strong potential in HIS processing, with a range of models being explored. However, the fixed geometric structure of convolutional kernels in the CNN model impedes interactions between long-term dependencies, and existing transformer methods have not fully harnessed the spectral structural characteristics of HSI data. To address these issues, we proposed a multiscale sample transformer (MSST) that effectively extracts spatial and spectral features via two separate branches. In the spectral branch, the spectral sequence of the individual target pixel is used as an input sample, and a spectral transformer (SpecFormer) is proposed to comprehensively extract the spectral structural features. In the spatial branch, a 3-D tensor consisting of target pixel and their neighboring pixels is used as an input sample, dimension reduction is performed, and a spatial transformer (SpatFormer) is designed to efficiently extract the deep spatial features of HSI data. Consequently, MSST enlarges the receptive field and captures the comprehensive joint spatial–spectral features. Extensive experiments conducted on five widely used HSI datasets demonstrate the superior classification performance of the proposed framework. Our source codes will be available at https://github.com/zhwt-xidian/Multi-Scale-Samples-Transformer_MSST**

***Index Terms*—Convolution neural network (CNN), hyperspectral image (HSI) classification, spatial–spectral transformer (SpecFormer).**

I. Introduction

REMOTE sensing technology is a comprehensive technique that utilizes the principles of electromagnetic waves to collect and process information from distant targets, enabling the detection and identification of various ground scenes [1]. Hyperspectral imaging captures numerous narrow spectral bands, providing rich spatial and spectral information for diverse applications such as land cover analysis, environmental monitoring, and material identification. Considering the distinctive characteristics of hyperspectral image (HSI) data, the designation of an efficient model is pivotal in addressing the HSI classification challenge, which serves as the fundamental basis for remote sensing applications, such as precision agriculture [2], [3], environmental monitoring [4], and mineral mapping [5], [6].

Traditional machine learning methods have been proposed in the very beginning for HSI classification. It usually comprises three separate processes: original feature extraction [7], [8], [9]; feature dimensionality reduction [10], [11]; and classifier design and training (if necessary). In the past decade, various advanced manual and data-based feature extraction methods emerged for HSI classification, such as independent component analysis (ICA) [12] and discrete wavelet transformation (DWT) [13]. Furthermore, spectral dimensionality reduction techniques, such as principal component analysis (PCA) [14] and linear discriminant analysis (LDA) [15], have been employed to enhance the efficiency of HSI classification. Classification model plays a crucial role in HSI classification, and the widely used classifiers include K-nearest neighbors [16], support vector machines (SVMs) [17], random forests [18], and so on. Although traditional machine learning methods can achieve excellent performance in certain specific classification tasks [19], they suffer from low generality and limited representation capability, hampering their ability to fully exploit the latent information within the data and resulting in limited classification accuracy.

In recent years, numerous deep learning-based methods have been proposed to address these challenges and harness the powerful feature learning capabilities of neural networks for HSI classification. Prominent deep learning approaches include deep belief networks (DBNs) [20], stacked autoencoders (SAEs) [21], convolutional neural networks (CNNs) [22], [23], [24], recurrent neural networks (RNNs) [25], [26], and full tensor networks [27]. Chen et al. [28] proposed the MH-2D-3D-CNN model, which combines the strengths of 2D-CNN and 3D-CNN within a multichannel framework. By incorporating techniques such as NAMD and CBAM for enhanced spectral–spatial feature extraction, the model addresses the challenges posed by small training sample sizes, achieving improved classification accuracy through dynamic feature learning. Liu et al. [29] proposed the sequential spectral–spatial feature convolution network (S^3FCN), which leverages a cascaded structure to sequentially extract spectral and spatial features. It introduces a novel spectral–spatial attention mechanism that enhances the integration of cross-domain features while maintaining computational efficiency. Chang et al. [30] proposed the IRTS-3DCNN model, which integrates the power of CNN with an iterative random training sampling strategy, enabling the reselection of training samples

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in each iteration and thereby enhancing feature learning and classification accuracy. These methods integrate the feature learning and classification tasks of the network in an end-to-end fashion, resulting in enhanced classification performance and allowing practitioners to focus on network design. However, these deep learning-based HSI classification methods still exhibit certain limitations. As a mainstream learning framework, CNN utilizes convolutional kernels with fixed geometric structures. Although it is able to grasp spatial features and local contextual relationships in images, the receptive field of the convolutional kernel is limited to a local scope, thus failing to capture interactions between long-range pixels in the feature maps [31]. However, various visual tasks involving spatial recognition have highlighted the significance of long-range pixel interactions across different locations. Furthermore, CNNs tend to overly emphasize the local spatial information in images, while paying limited attention to the inherent spectral information in HSI data. Yet, the spectral details play a vital role in classification tasks. In cases of spectral similarity between different materials, CNNs may struggle to effectively distinguish subtle spectral differences.

Recently, the transformer was initially proposed for machine translation and has become the state-of-the-art method for numerous natural language processing (NLP) tasks [32]. Taking inspiration from its success in the NLP domain, the transformer has rapidly found applications in image classification and remote sensing tasks as well. For example, Dosovitskiy et al. [33] proposed the vision transformer (ViT), which first divides the input image into patches and then employs position encodings to capture global sequence information. However, ViT is originally designed for RGB images, and applying it directly to HSI classification would demand substantial computational resources. Hong et al. [34] proposed a transformer-based backbone network, which utilizes the groupwise spectral embedding (GSE) module to capture local spectral information from neighboring spectral bands. Furthermore, it utilizes the cross-layer adaptive fusion (CAF) module to reduce information loss during data transmission between layers. The experimental results validate that the SpectralFormer surpasses CNNs in HSI classification tasks. However, SpectralFormer fails to fully extract joint spatial–spectral features. Zhong et al. [31] presented a spectral–spatial transformer (SpatFormer) with spatial attention and spectral association modules, integrating spectral and spatial information through spectral position and spatial position embeddings, respectively. However, the spectral association module primarily emphasizes the correlation between spectral information and spatial location, without considering the spectral structural features. Yang et al. [35] proposed the HiT model, the spectral-adaptive 3-D convolution projection (SACP) module utilizes two spectral-adaptive 3-D convolution layers to generate local spatial–spectral information from hyperspectral images. Subsequently, depthwise convolution operations are utilized to encode this spatial–spectral information across the dimensions of height, width, and spectrum. Sun et al. [36] proposed the SSFTT model, which utilizes a hybrid approach combining convolutional and transformer architectures for spectral–spatial feature extraction. Furthermore, it incorporates a Gaussian-weighted feature tokenizer to enhance semantic representation. Meng et al. [37] proposed the global–local multigranularity transformer (GLMGT) model, which combines the strengths of CNN and transformers to capture both global and local spectral–spatial features comprehensively. The model employs multigranularity feature extraction blocks and a gated feedforward module to effectively integrate fine- and coarse-grained features. Jiang et al. [38] presented the GraphGST model, which abandons CNNs and builds an absolute positional encoding (APE) to encode the absolute spatial sequences of pixels, thereby providing physically meaningful positional cues for the transformer. A self-supervised generative mechanism then performs cross-view contrastive learning to reinforce local-to-global representations. Other similar works specifically designed for HSI classification can be found in the literature [39], [40]. The spectral signature in HSI inherently represents the unique absorption patterns of land cover materials across different wavelengths. Precise extraction of these discriminative spectral features is essential for accurate classification, particularly when distinguishing spectrally similar materials. While current transformer-based approaches effectively capture spectral correlations between a target pixel and its neighborhood, they largely overlook the critical intrapixel spectral relationships—the inherent band-to-band dependencies within each pixel's spectral profile that encode vital contextual semantic information.

To overcome these limitations, this article presents a novel classification model called multiscale sample transformer (MSST). The MSST model aims to enhance the classification performance by addressing the absence of spectral structural similarity of transformer-based methods. We recommend to use two samples of different spatial scales for the classification of one certain pixel. Correspondingly, MSST incorporates two branches, namely, SpatFormer and spectral transformer (SpecFormer), which are responsible for extracting spatial and spectral features, respectively. Both branches employ the attention mechanism in different ways. The main contributions of this study are summarized as follows.

1) We propose a dual-scale sampling strategy in which two samples of varying spatial sizes are utilized for the classification of each target pixel. Samples with different scales focus on extracting spatial and spectral features of HSI data, thereby contributing to enhanced feature fusion (FF). We demonstrate the effectiveness of the proposed MSST through a series of ablation experiments and comparative trials.
2) In order to evaluate the spectral similarity of target pixel itself between multibands, we proposed a spectral sampling module (SSM), and a SpecFormer branch is built to extract more spectral structural details. Thus, it enhanced the model's ability to discriminate land cover objects with similar spectral characteristics. The effectiveness is verified through feature clustering experiments.

The rest of this article is organized as follows. Section II presents a brief illustration of the proposed MSST, including the SpecFormer branch and the SpatFormer branch. Section III illustrates HSI datasets, experimental settings, model settings,

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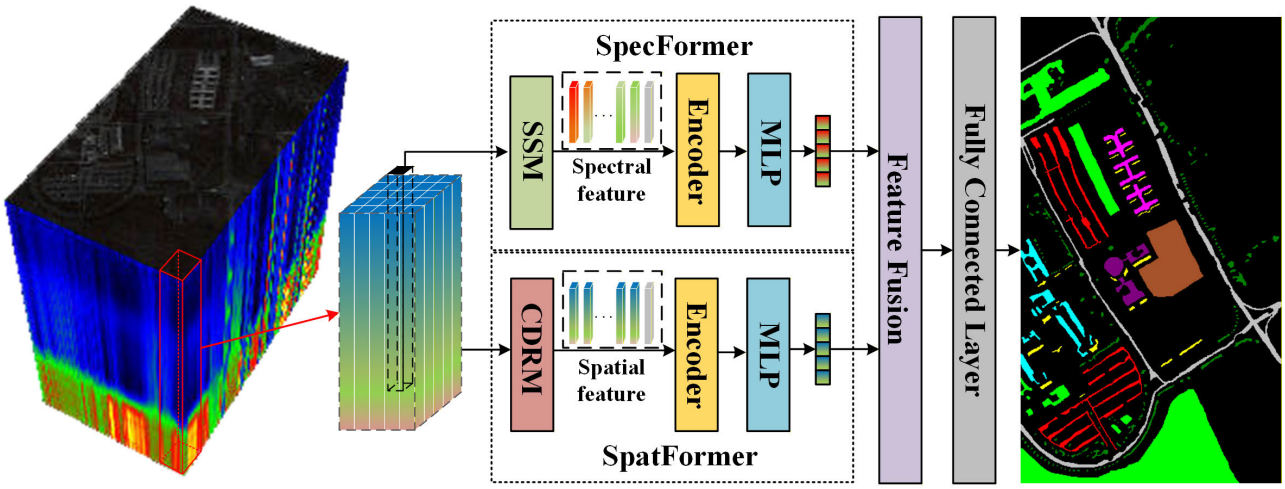


Fig. 1. Network architecture of MSST.

and experimental results. Finally, conclusions are given in Section IV.

II. Proposed Methodology

A. Overall Framework of MSST

In this article, we propose a spatial–spectral correlated multiscale sample classification network comprising dual branches for spatial and spectral features. The network effectively integrates complementary information from samples at different scales, thereby overcoming the limitation of missing spectral structural features in conventional transformer-based methods and improving the discrimination ability for land cover objects.

As depicted in Fig. 1, the overall architecture begins with two parallel branches processing the input data. The SpecFormer branch takes as input the spectral sequence of a target pixel extracted by the SSM, while the SpatFormer branch accepts a 3-D tensor—composed of the target pixel and its neighboring pixels—generated by the convolutional dimension reduction module (CDRM). The detailed descriptions of SSM and CDRM will be provided in Sections II-B and II-C, respectively. Inspired by the class token mechanism in BERT [41], a learnable class embedding is prepended to the input of each branch before being passed to the respective encoder. Since the features generated by SSM and CDRM lack positional information, we incorporate positional encoding into all input sequences to enhance the attention mechanism's sensitivity to positional relationships. The resulting sequences, now enriched with positional information, are fed into the encoder.

As illustrated in Fig. 2, within each encoder layer, the input samples are first normalized and then processed by either a spectral self-attention mechanism (SpecSAM) or a spatial self-attention mechanism (SpatSAM). The output of the attention mechanism is added to the original input via a residual connection. This sum is then normalized again, and the result is combined with the previous summation output to form the final output of the encoder layer. Each encoder layer maintains consistent input and output dimensions, allowing the output from one layer to be directly passed as input to the next.

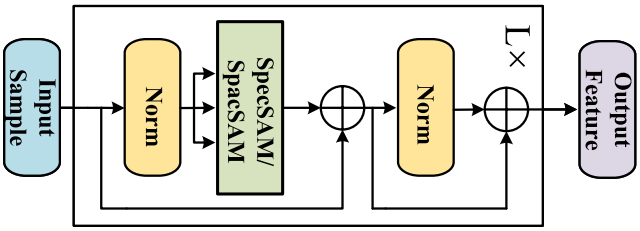


Fig. 2. Operation process of an encoder.

This process is repeated L times, with L denoting the number of encoder layers, which is kept the same for both SpecFormer and SpatFormer branches.

After the abovementioned processing, the class embeddings are extracted from the output of both encoders after L layers. These embeddings are then separately input into two fully connected (FC) layers to obtain the final extracted spectral and spatial features. Finally, the MSST model passes the extracted spectral and spatial features through the FF layer for FF and then into the FC layer to obtain the classification results.

B. Proposed SpecFormer

HSI captures spectral information from various positions, revealing the reflection characteristics of land targets at different wavelengths. Accurate and precise classification of targets in HSI scenes relies on effectively capturing the local reflection characteristics within this spectral information. To fully extract the meaningful representations from the spectral sequences, we introduce the SpecFormer branch as follows.

Fig. 3 illustrates the two components of SpecFormer, i.e., SSM and SpecSAM. The spectral information of the target pixel is summarized in a sequence with a channel size of c. SSM divides the target sequence into n short sequences of length c' through overlapping sampling, with the partitioning formula as follows:

$$c = (n-1)\,q + c', \quad c' \geq q \tag{1}$$

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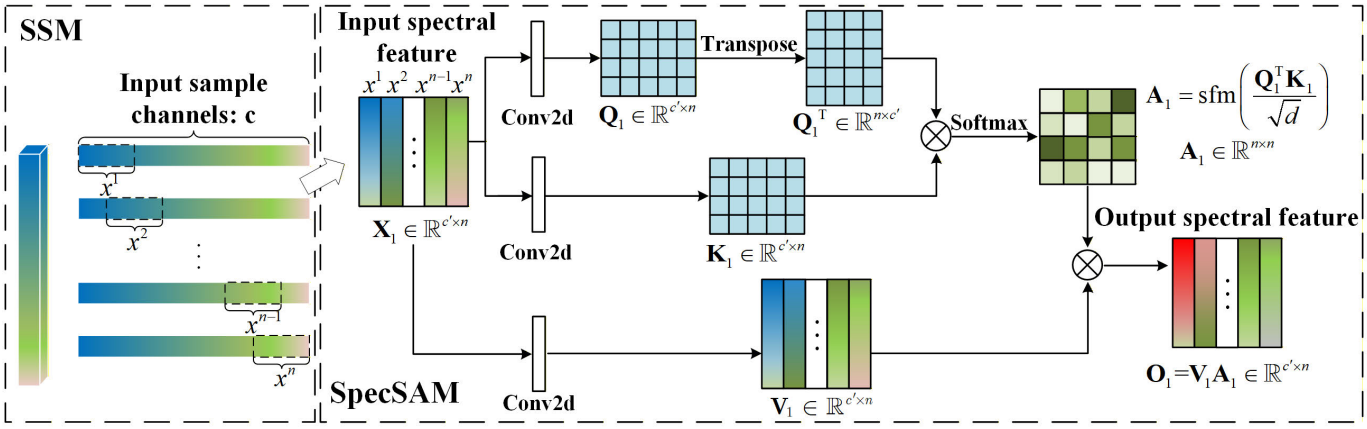


Fig. 3. SpecFormer branch.

where q represents the interval between starting points of adjacent short sequences. It is noteworthy to mention that the partitioning process must satisfy condition $c' \geq q$ to avoid any loss of important spectral information, which could adversely impact the classification results. During the experimentation phase, we conduct various tests with different combinations of sequence numbers n and sequence lengths c' and carefully select the combination that produces the best evaluation metrics as the optimal parameters for the model on each dataset.

SSM divides the sample sequence into n shorter sequences, creating the input spectral feature vector group $\mathbf{X}_1 = \{\mathbf{x}^i\} \in \mathbb{R}^{c' \times n}$. SpecSAM serves as the key component of the encoder in the SpecFormer branch, enhancing the SpecFormer's feature extraction ability by emphasizing crucial features in the spectrum while reducing the impact of irrelevant ones. Specifically, 2-D convolutions with a kernel size of 1×1 (1×1 conv) are employed to construct the essential query, key, and value components. The specific calculations can be expressed as follows:

$$\mathbf{Q}_1 = F\left(\mathbf{X}_1, \mathbf{W}_q\right), \quad \mathbf{Q}_1 \in \mathbb{R}^{c' \times n}, \quad \mathbf{W}_q \in \mathbb{R}^{c' \times c'} \tag{2}$$

$$\mathbf{K}_1 = F\left(\mathbf{X}_1, \mathbf{W}_k\right), \quad \mathbf{K}_1 \in \mathbb{R}^{c' \times n}, \quad \mathbf{W}_k \in \mathbb{R}^{c' \times c'} \tag{3}$$

$$\mathbf{V}_1 = F\left(\mathbf{X}_1, \mathbf{W}_v\right), \quad \mathbf{V}_1 \in \mathbb{R}^{c' \times n}, \quad \mathbf{W}_v \in \mathbb{R}^{c' \times c'} \tag{4}$$

where $\mathbf{W}_q$, $\mathbf{W}_k$, and $\mathbf{W}_v$ are the weight matrices for the query $\mathbf{Q}_1$, key $\mathbf{K}_1$, and value $\mathbf{V}_1$, respectively. $F(\cdot)$ denotes the 1×1 convolution. The formula for calculating the attention scores $\mathbf{A}_1$ is as follows:

$$\mathbf{A}_1 = \text{softmax}\left(\frac{\mathbf{Q}_1^{\mathrm{T}} \mathbf{K}_1}{\sqrt{d}}\right), \quad \mathbf{A}_1 \in \mathbb{R}^{n \times n} \tag{5}$$

where d represents the dimension of the query and key vectors. Finally, the weight matrix $\mathbf{V}_1$ is multiplied by the obtained attention scores $\mathbf{A}_1$ to reconstruct the values at each position of the output spectral features

$$\mathbf{O}_1 = \mathbf{V}_1 \mathbf{A}_1, \quad \mathbf{O}_1 \in \mathbb{R}^{c' \times n}. \tag{6}$$

C. Proposed SpatFormer

In HSI classification, the input data contain hundreds of spectral bands, providing rich spectral information but also introducing the challenge of the "curse of dimensionality" [41]. Conventional transformer-based models are particularly affected by high-dimensional inputs in two main aspects. First, the self-attention mechanism scales quadratically with input dimension, leading to a rapid growth in parameter count and computational cost, which in turn hinders classification performance. Second, hyperspectral bands are often highly correlated and contain substantial redundancy. Feeding such high-dimensional data directly into the self-attention layer impedes the model's ability to distinguish relevant features from redundant information, thus limiting classification accuracy.

To overcome these limitations, we propose the SpatFormer architecture. As shown in Fig. 4, SpatFormer consists of two key components: the CDRM and the SpatSAM. In the CDRM, 1×1 convolution is first applied to reduce input dimensionality and extract discriminative features. Unlike conventional PCA, 1×1 convolution incorporates nonlinear transformations, which help capture more complex and nonlinear relationships in the data. Moreover, it can be seamlessly integrated into the transformer framework, enabling end-to-end HSI classification. The CDRM process is formulated as follows:

$$\begin{aligned} &\boldsymbol{\mathcal{X}}_2' = \text{ReLU}\left(\text{BN}\left(F\left(\boldsymbol{\mathcal{X}}_2, \mathbf{W}_F\right)\right)\right) \\ &\boldsymbol{\mathcal{X}}_2 \in \mathbb{R}^{c \times h \times w}, \quad \boldsymbol{\mathcal{X}}_2' \in \mathbb{R}^{c' \times h \times w} \end{aligned} \tag{7}$$

where $\boldsymbol{\mathcal{X}}_2$ denotes the input spatial sample; c stands for the number of channels; and h and w refer to height and width, respectively. $F(\cdot)$ represents the 1×1 conv, and $\mathbf{W}_F$ is the weight matrix for the convolution operation. $\text{BN}(\cdot)$ stands for batch normalization, and $\text{ReLU}(\cdot)$ represents the rectified linear unit activation function.

SpatSAM serves as the key component in the encoder of the SpecFormer branch, aiming to simulate the interactions among different spatial samples in HSI data. By overcoming the constraints of the convolutional kernel structure, SpatSAM enhances the feature extraction capacity while reducing the computational cost for handling large spatial size samples. The specific computation process is as follows:

$$\mathbf{Q}_2 = F\left(\boldsymbol{\mathcal{X}}_2', \quad \mathbf{W}_q\right), \quad \mathbf{Q}_2 \in \mathbb{R}^{c' \times \text{hw}}, \quad \mathbf{W}_q \in \mathbb{R}^{c' \times c'} \tag{8}$$

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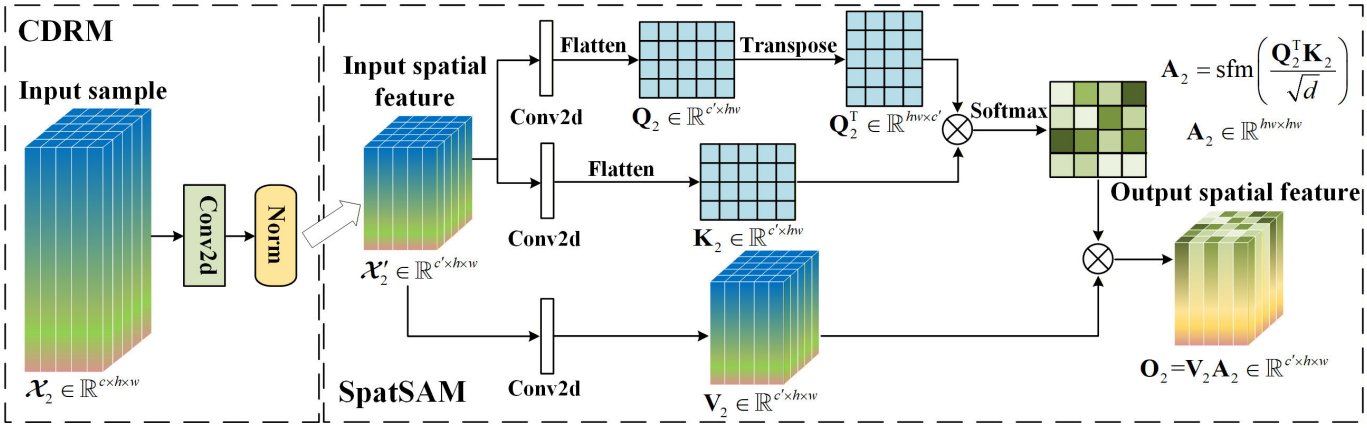


Fig. 4. SpatFormer branch.

$$\mathbf{K}_2 = F\left(\mathcal{X}'_2, \quad \mathbf{W}_k\right), \quad \mathbf{K}_2 \in \mathbb{R}^{c' \times \mathrm{hw}}, \quad \mathbf{W}_k \in \mathbb{R}^{c' \times c'} \tag{9}$$

$$\mathbf{V}_2 = F\left(\mathcal{X}'_2, \quad \mathbf{W}_v\right), \quad \mathbf{V}_2 \in \mathbb{R}^{c' \times \mathrm{hw}}, \quad \mathbf{W}_v \in \mathbb{R}^{c' \times c'}. \tag{10}$$

Here, similar to the previously introduced SpecSAM, the weight matrices $\mathbf{W}_q$, $\mathbf{W}_k$, and $\mathbf{W}_v$ represent the parameters for the query $\mathbf{Q}_2$, key $\mathbf{K}_2$, and value $\mathbf{V}_2$, respectively. $F(\cdot)$ represents the 1×1 conv operation. The attention scores $\mathbf{A}_2$ are calculated using the following formula:

$$\mathbf{A}_2 = \mathrm{softmax}\left(\frac{\mathbf{Q}_2^{\mathrm{T}}\mathbf{K}_2}{\sqrt{d}}\right), \quad \mathbf{A}_2 \in \mathbb{R}^{\mathrm{hw} \times \mathrm{hw}} \tag{11}$$

where d represents the dimension of the query and key vectors. Finally, the output of SpatSAM can be calculated as follows:

$$\mathbf{O}_2 = \mathbf{V}_2\mathbf{A}_2, \quad \mathbf{O}_2 \in \mathbb{R}^{c' \times h \times w}. \tag{12}$$

Here, the value at each position of the output feature $\mathbf{O}_2$ is the summation reconstruction of the tensor $\mathbf{V}_2$ weighted by the attention scores $\mathbf{A}_2$.

D. FF and Classification

After the feature extraction process, it effectively combines the spectral and spatial features to further improve the accuracy of HSI data classification tasks. In this step, we utilize a data concatenation approach to merge the features extracted from both branches. The fused spatial–spectral features are subsequently fed into the FC layer, resulting in the final classification results. This process can be summarized as follows:

$$\mathbf{O} = \mathrm{FC}\left(\mathrm{Cat}_c\left(\mathrm{BN}(\mathbf{O}_{s1}), \mathrm{BN}(\mathbf{O}_{s2})\right)\right)$$
$$\mathbf{O} \in \mathbb{R}^c, \ \mathbf{O}_{s1}, \mathbf{O}_{s2} \in \mathbb{R}^c. \tag{13}$$

Among them, $\mathrm{Cat}_c(\cdot)$ refers to channel-wise concatenation; $\mathrm{FC}(\cdot)$ represents the fully connected layer; $\mathrm{BN}(\cdot)$ denotes batch normalization; $\mathbf{O}_{s1}$ and $\mathbf{O}_{s2}$ are the outputs of SpecFormer and SpatFormer, respectively; and $\mathbf{O}$ is the final output of MSST. Here, c denotes the number of classes in the dataset.

III. Experiments

To validate the effectiveness of the proposed MSST model for HSI classification, we conduct experiments on five well-known datasets: the Salinas (SA) dataset, Pavia University (PU) dataset, Indian Pines (IPs) dataset, Houston2013 (Hou13) dataset, and Houston2018 (Hou18) dataset. We present brief descriptions of these datasets and explore how different hyperparameter settings affect the network's performance. Additionally, we conduct extensive ablation experiments and compare MSST's classification results with other state-of-the-art methods, evaluating its performance on HSI data quantitatively and qualitatively.

A. HSI Datasets

They are given as follows.

1) *PU Dataset:* The PU dataset consists of spectral images acquired by the RoSIS spectrometer at the University of Pavia, Italy, in 1992. The dataset includes images with a size of 610×340 pixels and comprising a total of 115 spectral bands with a spatial resolution of 1.3 m. For the experiments, 103 selected spectral bands from this dataset were used.
2) *SA Dataset:* This dataset was collected by the Airborne Visible Imaging Spectrometer (AVIRIS) sensor as it flew over the SA Valley in California, USA, with a spatial resolution of 3.7 m/pixel. The scene comprises a total of 224 spectral bands, and after removing 20 water-absorbing bands (108–112, 154–167, and 224), it consists of 512×217 pixels and 204 spectral bands. Apart from the unclassified pixels, the dataset includes 16 manually labeled classes.
3) *IPs:* This dataset was obtained in 1992 using the AVIRIS sensor in North-Western Indiana, USA. The HSI consists of 145×145 pixels in the spatial dimension and 220 bands in the spectral dimension. There are 200 spectral bands retained after removing 20 water-absorption bands. Apart from the unclassified pixels, the dataset includes 16 manually labeled classes. Unlike the other datasets mentioned above, this dataset is an imbalanced-class dataset.
4) *Hou13 Dataset:* This dataset was acquired in 2013 over the University of Houston campus and its surrounding urban areas using the ITRES CASI-1500 airborne sensor. The HSI comprises 349×1905 pixels in the spatial dimension and 144 spectral bands. The dataset includes

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15 manually annotated land cover classes, excluding unlabeled pixels, and exhibits a relatively balanced class distribution compared to other datasets.

5) *Hou18 Dataset:* This dataset was acquired in 2018 over the University of Houston campus and its neighboring urban area by the National Center for Airborne Laser Mapping (NCALM). The HSI consists of 1202×4768 pixels in the spatial dimension and 48 spectral bands. The dataset provides 20 finely annotated land cover classes, excluding the unlabeled background (BKG), and is characterized by a significant class imbalance, with certain categories (such as healthy grass and residential buildings) dominating the scene.

B. Experimental Settings

To ensure a fair comparison, all experiments are conducted in the same environment. Specifically, an Intel Core i7-12700H CPU and an NVIDIA RTX 3060 GPU are used. The operating system is Windows 11, and the experimental software environment consists of CUDA 12.2, Python 3.8.2, and PyTorch 1.9.0. During the training phase, a smaller learning rate of 0.001 is chosen to mitigate the risk of convergence to local optima. The Adam optimizer is employed for model optimization, and the cross-entropy loss function is used for backpropagation. To quantitatively evaluate the performance of the proposed method, we use various metrics, including overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ). These metrics provide comprehensive assessments of the classification performance.

To achieve optimal network performance, we conduct comparative experiments to fine-tune the model's performance on each dataset. To validate the effectiveness of our proposed method for HSI classification, we perform comparative experiments on the PU, SA, IP, Hou13, and Hou18 datasets, where we compare the MSST method against other approaches, including the state-of-the-art SAE-CNN [21], 3D-CNN [24], IRTS-3DCNN [30], the transformer-based methods ViT [33] and SpectralFormer [34]; the hybrid CNN-transformer architectures, including HiT [35] SSFTT [36]; and graph generative structure-aware transformer GraphGST [38]. The detailed configurations of these compared methods are as follows.

1) *For SAE-CNN:* The input data are reduced to five dimensions using their respective dimension reduction models as described in their original articles. SAE-CNN consists of four 2-D convolutional blocks, one average pooling layer, and one softmax layer. Each 2-D convolutional block in the CNN is composed of a 2-D convolutional layer, a batch normalization (BN) layer, and a ReLU activation function.
2) *For 3D-CNN:* It comprises four 3-D convolutional blocks and one softmax layer. Except for the last 3-D convolutional block, the other three 3-D convolutional blocks are composed of a 3-D convolutional layer, a ReLU activation function, and a max-pooling layer.
3) *For IRTS-3DCNN:* We followed the network architecture and training methodology outlined in the original article. Additionally, the input size is set to 9×9, and the tanimoto index (TI) is configured to 0.99.
4) *For SpectralFormer:* The input is configured with a patch-wise approach, where each patch size is set to 7×7, and the band patches are set to 3. The model follows the methodology in the original article, utilizing five encoder layers.
5) *For ViT:* The input size is set to 9×9, and the patch size is set to 3×3. To ensure fairness, ViT uses the same configuration of five encoder layers as SpectralFormer.
6) *For HiT:* We follow the architecture described in their original articles.
7) *For SSFTT:* The input size is set to 9×9, the number of feature tokens is set to 5, and the number of encoder layers is set to 5.
8) *For GraphGST:* The input patch size is set to 5×5, and a sliding window of size 9×9 is adopted to extract local spatial–spectral features. A threshold value of 0.2 is applied to sparsify the constructed graph. GraphGST employs three transformer encoder layers, consistent with the baseline architectures.
9) *For MSST:* The spectral sample size is set as a single pixel, and the spatial sample size is 9×9. Similar to SpectralFormer and ViT, both branches of MSST use five encoder layers. In detail, the SpecFormer branch utilizes SSM for overlapping sampling of spectral samples. For the PU dataset, the spectral length c' is 59, and the number n of sequences is 5. For the SA dataset, c' is 52 and n is 5. For the IP dataset, c' is 152 and n is 3.

To avoid the randomness of experimental results, each experiment is repeated ten times, and each time the experiment is randomly sampled. The average results from ten independent runs are reported as the final experimental outcomes. In Tables I–IX, the standard deviation is provided in parentheses next to each result. In this article, the best experimental results in the tables are highlighted in bold.

C. Model Settings

1) Different Spectral Input: SpecFormer takes single-pixel spectral information as input, and SSM performs overlapping sampling on the spectral data. If the number of sampled spectral sequences is too large and the spectral length is too short, it may result in increased computational complexity and information loss. This is due to the fact that short spectral lengths can lead to the truncation or loss of important spectral features. Conversely, if the number of sampled spectral sequences is too small and the spectral length is too long, the model might miss capturing subtle spectral variations and differences, which can negatively impact the classification accuracy. Therefore, finding the right balance in the sampling parameters is crucial for ensuring optimal performance in the SpecFormer branch.

In order to determine the optimal sampling spectral length c' and quantity n for SpecFormer, a series of experiments is conducted on the PU and SA datasets. Since different datasets have varying numbers of spectral bands, different values of c' and n are selected for comparison experiments based on Formula (1). In each experiment, 50 random points are selected as sample points for each class, while all other experimental parameters remain constant as previously described.

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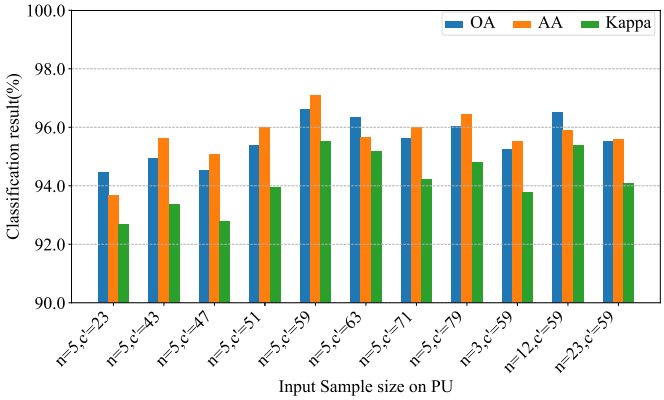


Fig. 5. Classification result under different input spatial sizes on PU.

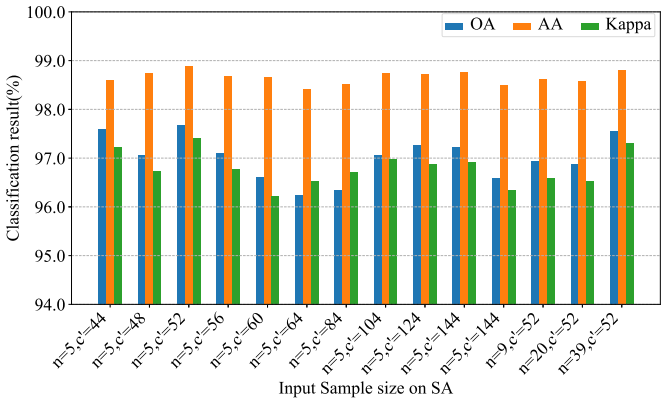


Fig. 6. Classification result under different input spatial sizes on SA.

The experimental results for the PU dataset are shown in Fig. 5, where the first eight sets of experiments kept the n fixed at 5 while changing c' from 23 to 79. The comparison results indicate that the SpecFormer achieves the best classification performance when $n = 5$ and $c' = 59$, with the OA value of 96.60%, AA value of 97.09%, and κ value of 95.52%. Subsequently, we conduct three additional control experiments, keeping c' fixed at 59 and varying n from 3 to 23. The results demonstrate that among all parameter combinations, selecting $n = 5$ and $c' = 59$ results in the best classification performance. Therefore, it is recommended to use the input combination of $n = 5$ and $c' = 59$ for the PU dataset.

We conduct similar experiments on the SA dataset, and the results are shown in Fig. 6. Among all the experiment results, the model achieves the best classification performance when $n = 5$ and $c' = 52$, with the OA value of 97.68%, AA value of 98.88%, and κ value of 97.42%. It is worth noting that when we select $n = 5$ and $c' = 52$, the model still achieves a relatively good classification performance, with the OA value of 97.56%, AA of 98.81%, and κ value of 97.30%, which is close to the best result. However, compared to $n = 5$, using $n = 39$ leads to a higher number of parameters, increasing the training cost of the network. Therefore, we recommend selecting $n = 5$ and $c' = 52$ as the input combination for the SA dataset. When applying the MSST model to classify other datasets in the future, it is essential to choose appropriate spectral input dimensions based on the specific spectral characteristics of each dataset.

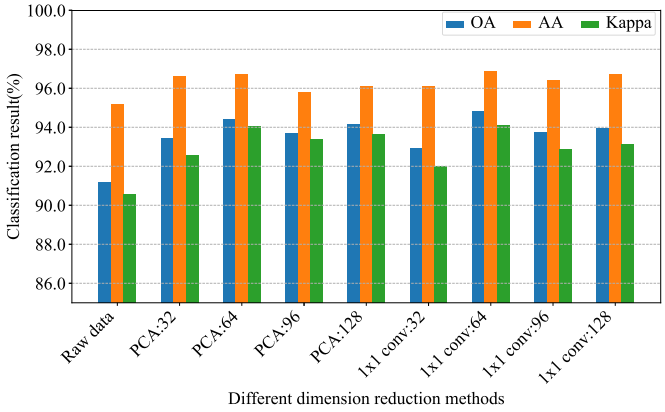


Fig. 7. Dimension reduction results on IP.

2) Different Dimensionality Reduction Method: SpatFormer takes a 3-D tensor as input, which consists of the target pixel and its neighboring pixels. HSI data usually have high spectral dimensionality, containing redundant information. Moreover, when the input data have high dimensions, the self-attention matrix of the transformer significantly expands, resulting in an increased number of network parameters and degraded classification performance. Therefore, an appropriate dimension reduction scheme is crucial for the SpatFormer branch. To investigate the classification performance on different dimension reduction schemes and find out an appropriate target dimensionality c', we conducted experiments on the IP dataset using two schemes: PCA and 1×1 Conv, and then compared the results across four target dimensions: 32, 64, 96, and 128.

As depicted in Fig. 7, compared to the input raw data, dimensionality reduction significantly improves the classification results. Among all the dimensionality reduction results, reducing the data to dimension $c' = 64$ using 1×1 conv achieves the best classification performance. This is because 1×1 conv learns its dimensionality-reduced representation end-to-end for the specific classification task, whereas PCA employs a fixed, task-agnostic projection based solely on the data.

3) Different Input Spatial Sizes: In HSI, each pixel contains valuable information from multiple spectral bands. If the input spatial size is too small, important spatial details may be overlooked, leading to a decline in classification accuracy. Conversely, an overly large input spatial size may cause the model to overfit to noise in the training data, resulting in reduced generalization performance. Additionally, a larger input spatial size will increase computational costs, leading to longer training times and higher demands for computing resources. Thus, selecting an appropriate input spatial size is crucial for achieving optimal classification performance.

To determine the optimal input spatial size for the MSST method, a series of experiments is conducted on the PU dataset and SA dataset. We evaluate four different input spatial sizes: 5×5, 7×7, 9×9, and 11×11. For each category, 50 random

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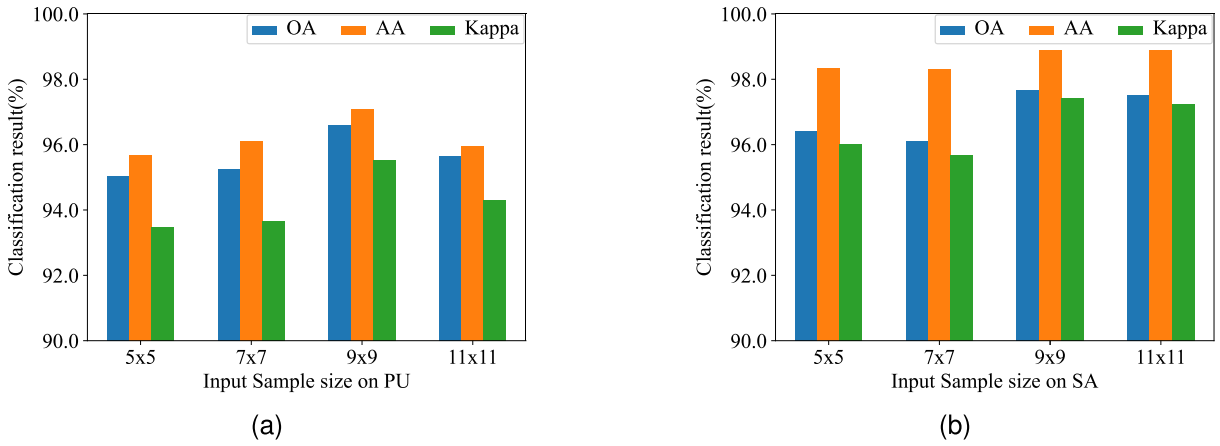


Fig. 8. Classification result under different input spatial sizes. (a) PU dataset. (b) SA dataset.

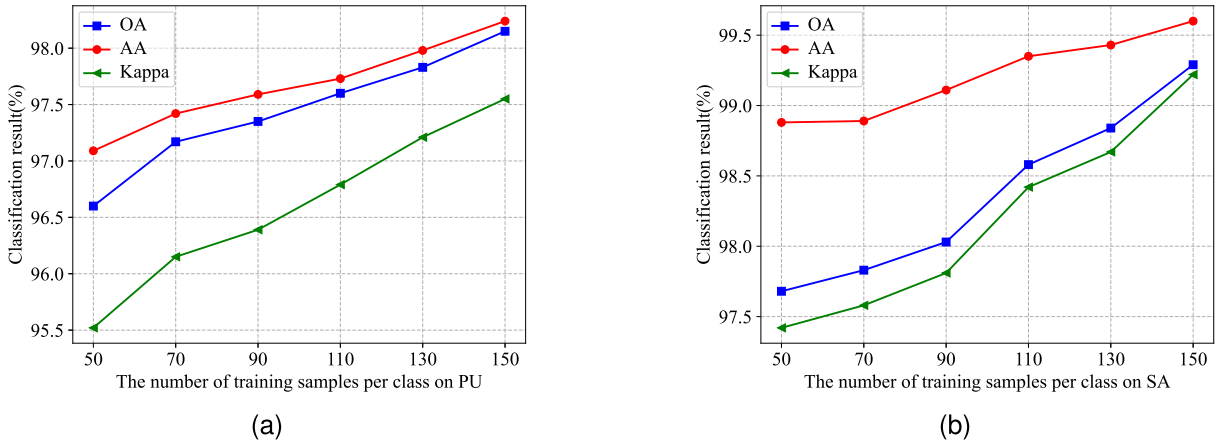


Fig. 9. Classification result under different numbers per class. (a) PU dataset. (b) SA dataset.

points are selected as sample points while keeping all other experimental parameters unchanged.

As depicted in Fig. 8, the MSST model demonstrates a significant advantage across all three evaluation metrics for the PU dataset when utilizing an input spatial size of 9×9. For the SA dataset, the MSST model does not show as significant improvement as in the PU dataset, but it still maintains a slight advantage in all evaluation metrics. These disparities arise from the distinctive spatial distribution of land cover in each dataset. Therefore, it is recommended to adopt a 9×9 input spatial size for both the SA and PU datasets in future experiments. When applying the MSST model to other datasets for classification, prioritizing the use of a 9×9 input spatial size is suggested, but it is also advisable to try other appropriate input spatial sizes based on the specific spatial distribution characteristics of the land cover within the dataset.

4) Different Numbers per Class: In deep learning models for HSI classification, having a sufficient number of labeled samples is essential for network optimization. However, accurately labeling pixels in HSI data is challenging in practice. Thus, HSI classification deep learning models should be able to adapt to variations in the number of training samples. To explore the classification performance of MSST method with varying training sample sizes, we conduct experiments on the PU and SA datasets, increasing the number of training samples per class from 50 to 150 with intervals of 20 while keeping all other experimental parameters unchanged. As depicted in Fig. 9, the performance curves of MSST remain relatively stable and gradually improve as the number of training samples increases. This indicates that the proposed method demonstrates good adaptability to changes in the training sample size, providing consistent improvements in various evaluation metrics with the increase in training data.

D. Experimental Results

1) Ablation Study: In order to validate the effectiveness of MSST in improving classification performance by extracting joint spectral–spatial features from different input scales, ablation experiments are conducted in this section. The experiments were performed on the PU dataset and SA dataset using SpecFormer branch, SpatFormer branch, and MSST separately. For each class, 50 sample points are selected while keeping other experimental parameters constant. The classification results are presented in Table I.

Through the analysis of the results shown in Table I, several key observations can be made. SpecFormer, which integrates

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TABLE I
ABLATION ANALYSIS OF THE PROPOSED MSST WITH SPECFORMER AND SPATFORMER

Methods	SSM	CDRM	PU			SA		
			OA(%)	AA(%)	κ(%)	OA(%)	AA(%)	κ(%)
SpecFormer	✘		90.31(0.36)	90.43(0.34)	88.87(0.38)	92.34(0.30)	93.19(0.33)	92.17(0.32)
	✔		92.07(0.28)	92.67(0.27)	89.60(0.29)	94.91(0.26)	97.69(0.25)	94.35(0.28)
SpatFormer		✘	91.45(0.35)	91.37(0.33)	91.32(0.31)	93.57(0.29)	93.82(0.30)	93.14(0.31)
		✔	95.53(0.23)	95.59(0.22)	94.12(0.24)	97.68(0.20)	98.88(0.18)	97.42(0.21)
MSST	✔	✔	**96.60**(0.14)	**97.09**(0.13)	**95.52**(0.15)	**97.99**(0.11)	**99.10**(0.10)	**97.76**(0.12)

TABLE II
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE PU DATASET

Class No.	Train Samples	3D-CNN	SAE-CNN	IRTS-CNN	ViT	Spectral-Former	HiT	SSFTT	GraphGST	MSST
1	66	87.30(0.78)	90.17(0.70)	95.29(0.50)	89.16(0.73)	86.95(0.80)	94.51(0.52)	94.19(0.53)	90.90(0.15)	**98.73**(0.19)
2	186	93.47(0.33)	92.63(0.36)	97.68(0.20)	93.86(0.32)	94.80(0.30)	99.55(0.04)	98.80(0.09)	**99.82**(0.02)	99.73(0.03)
3	20	88.49(0.92)	89.06(0.91)	53.31(2.10)	**96.05**(0.46)	87.50(1.00)	89.89(0.89)	83.04(1.25)	90.00(0.37)	88.99(0.94)
4	30	97.88(0.22)	98.78(0.15)	93.31(0.55)	96.29(0.28)	97.09(0.24)	95.37(0.35)	96.44(0.30)	93.33(0.10)	**98.60**(0.13)
5	13	99.96(0.02)	99.99(0.01)	92.71(1.20)	99.93(0.03)	99.91(0.03)	98.36(0.20)	**100.0**(0.00)	**100.0**(0.05)	**100.0**(0.00)
6	50	94.75(0.40)	93.19(0.45)	91.91(0.50)	96.20(0.34)	97.51(0.28)	92.76(0.48)	91.13(0.52)	92.00(0.20)	**98.27**(0.25)
7	13	92.93(0.65)	95.38(0.55)	78.80(1.85)	99.22(0.08)	93.83(0.60)	70.37(2.60)	92.56(0.66)	**99.51**(0.06)	92.71(0.66)
8	36	83.25(0.85)	91.40(0.55)	87.51(0.70)	93.19(0.49)	83.06(0.89)	87.67(0.65)	84.95(0.81)	94.44(0.30)	**94.92**(0.38)
9	9	98.88(0.15)	99.77(0.03)	88.38(1.75)	98.08(0.18)	99.39(0.08)	97.99(0.22)	96.73(0.28)	**100.0**(0.02)	99.89(0.02)
OA(%)		92.16(0.20)	92.95(0.18)	92.32(0.21)	94.08(0.16)	92.93(0.18)	93.72(0.17)	94.85(0.14)	96.19(0.06)	**98.18**(0.08)
AA(%)		92.99(0.25)	94.49(0.21)	86.55(0.45)	95.77(0.18)	93.34(0.22)	88.50(0.39)	93.09(0.23)	95.63(0.10)	**96.87**(0.13)
κ(%)		89.74(0.23)	90.77(0.21)	89.80(0.24)	92.25(0.18)	90.75(0.21)	91.64(0.20)	93.16(0.17)	94.88(0.08)	**97.58**(0.10)
PreTime		12.6s	7.9s	12.5s	22.2s	21.4s	23.5s	16.2s	20.58s	16.8s

the SSM module and focuses solely on spectral structural features, lacks the incorporation of broader spatial contextual information, resulting in the lowest classification accuracy among the compared methods. In contrast, SpatFormer, which adopts the CDRM module and emphasizes spatial correlations, achieves better performance than SpecFormer, indicating that spatial information plays a more dominant role than spectral structure alone in HSI classification. MSST, as a fusion of SpecFormer and SpatFormer, incorporates both SSM and CDRM modules to jointly exploit spectral and spatial contextual information. This hybrid architecture leads to superior classification results across both PU and SA datasets. Specifically, on the PU dataset, MSST achieves an OA improvement of 4.53% over SpecFormer with SSM and 1.07% over SpatFormer with CDRM. Similarly, on the SA dataset, MSST surpasses SpecFormer with SSM by 3.08% and SpatFormer with CDRM by 0.31% in terms of OA. These improvements highlight MSST's strong capacity to capture and integrate features from multiple spatial scales and spectral dimensions, enabling more accurate land cover classification. These findings strongly suggest that MSST's excellent ability to extract features from different input scales. MSST utilizes the joint spectral–spatial features for accurate land cover classification. Based on the extraction of spatial features from SpatFormer, incorporating additional spectral structural details from SpecFormer contributes to enhancing classification performance.

2) Classification Result of Different Methods: To further validate the effectiveness and generalization ability of the proposed MSST, we compare it with other state-of-the-art methods, including the state-of-the-art SAE-CNN, 3D-CNN, and IRTS-3DCNN methods, as well as the transformer-based ViT, SpectralFormer, HiT, SSFTT, and GraphGST models. All network configurations are as previously described. The number of training samples selected for each dataset is presented in Tables II–VI, where the validation set contains half the number of training samples, and the remaining samples are used for testing. The corresponding classification results of the compared methods on different datasets are also shown in these tables.

First and foremost, when considering the evaluation metrics (OA, AA, and κ), MSST consistently outperforms the other methods, showcasing its superior classification performance. Specifically, compared to the second-best performing method on the PU dataset, MSST achieves an increase of 1.99% in OA, 1.24% in AA, and 2.70% in κ. On the SA dataset, MSST improves OA by 2.25%, AA by 1.73%, and κ by 2.52%. On the IP dataset, MSST improves OA by 0.35%, AA by 0.65%, and κ by 0.43%. On the Hou13 dataset, MSST improves OA by 0.26%, AA by 0.17%, and κ by 0.28%. On the Hou18

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TABLE III
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE SA DATASET

Class No.	Train Samples	3D-CNN	SAE-CNN	IRTS-CNN	ViT	Spectral-Former	HiT	SSFTT	GraphGST	MSST
1	20	99.81(0.02)	99.75(0.02)	99.90(0.01)	99.97(0.01)	99.97(0.01)	99.45(0.03)	99.99(0.01)	**100.0**(0.01)	99.99(0.01)
2	37	99.74(0.02)	99.54(0.03)	99.97(0.01)	99.94(0.01)	**99.99**(0.01)	99.73(0.02)	99.60(0.02)	99.98(0.01)	99.97(0.01)
3	20	99.23(0.05)	99.81(0.02)	99.97(0.02)	99.88(0.02)	99.91(0.02)	90.49(0.50)	98.99(0.08)	92.88(0.05)	**99.98**(0.01)
4	14	**99.96**(0.01)	99.78(0.02)	96.92(0.15)	99.63(0.03)	99.89(0.02)	99.78(0.02)	97.78(0.12)	99.32(0.01)	99.21(0.05)
5	27	99.02(0.06)	99.80(0.02)	97.83(0.10)	99.61(0.03)	99.93(0.02)	98.73(0.08)	99.37(0.04)	**100.0**(0.01)	**100.0**(0.00)
6	40	**100.0**(0.00)	99.95(0.01)	**100.0**(0.00)	99.99(0.01)	**100.0**(0.00)	**100.0**(0.00)	**100.0**(0.00)	94.74(0.05)	**100.0**(0.00)
7	36	98.83(0.08)	99.88(0.02)	**100.0**(0.00)	99.87(0.02)	99.67(0.03)	98.66(0.08)	**100.0**(0.00)	**100.0**(0.01)	99.88(0.02)
8	113	80.69(0.90)	84.13(0.80)	81.35(0.85)	86.11(0.75)	85.58(0.78)	94.23(0.22)	93.42(0.25)	94.64(0.12)	**96.93**(0.15)
9	33	98.69(0.08)	99.44(0.04)	99.89(0.02)	99.92(0.02)	99.79(0.03)	97.73(0.10)	99.98(0.01)	**100.0**(0.01)	99.84(0.02)
10	11	97.11(0.30)	97.46(0.28)	98.69(0.18)	97.86(0.25)	98.55(0.20)	98.51(0.20)	98.17(0.22)	87.50(0.15)	**99.66**(0.06)
11	19	99.34(0.05)	99.51(0.04)	97.75(0.12)	99.79(0.03)	**99.98**(0.01)	98.60(0.09)	97.38(0.14)	98.71(0.01)	98.97(0.06)
12	9	99.43(0.05)	99.97(0.01)	**100.0**(0.00)	99.85(0.03)	99.99(0.01)	99.84(0.02)	99.74(0.03)	**100.0**(0.01)	99.84(0.02)
13	27	99.82(0.02)	**99.98**(0.01)	83.84(0.80)	99.96(0.01)	**99.98**(0.01)	95.31(0.35)	99.45(0.04)	99.82(0.01)	99.78(0.02)
14	11	98.69(0.10)	99.20(0.06)	96.92(0.15)	99.55(0.03)	99.63(0.03)	91.31(0.50)	90.38(0.60)	**100.0**(0.01)	98.13(0.12)
15	73	86.25(0.75)	87.02(0.70)	92.16(0.55)	93.31(0.45)	94.56(0.35)	79.09(0.90)	82.69(0.80)	91.67(0.28)	**94.61**(0.35)
16	18	98.20(0.12)	**99.76**(0.02)	98.67(0.10)	99.46(0.04)	99.42(0.04)	92.31(0.45)	96.07(0.20)	98.63(0.01)	98.34(0.10)
OA(%)		93.50(0.35)	94.62(0.30)	94.36(0.32)	95.99(0.25)	96.09(0.24)	94.97(0.30)	96.27(0.22)	96.20(0.08)	**98.45**(0.12)
AA(%)		97.17(0.25)	97.81(0.22)	96.49(0.28)	98.42(0.18)	98.55(0.17)	95.98(0.30)	97.25(0.20)	97.34(0.06)	**99.07**(0.10)
κ(%)		92.78(0.38)	94.02(0.32)	93.73(0.34)	95.54(0.27)	95.66(0.26)	94.39(0.32)	95.84(0.24)	95.75(0.10)	**98.27**(0.14)
PreTime		21.0s	9.6s	20.8s	31.2s	35.0s	43.0s	20.1s	33.67s	25.6s

TABLE IV
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE IP DATASET

Class No.	Train Samples	3D-CNN	SAE-CNN	IRTS-CNN	ViT	Spectral-Former	HiT	SSFTT	GraphGST	MSST
1	25	99.93(0.03)	99.97(0.01)	97.83(0.10)	99.96(0.02)	99.91(0.04)	99.97(0.01)	99.95(0.01)	**100.0**(0.01)	99.98(0.01)
2	50	86.97(0.40)	73.25(0.50)	83.54(0.45)	80.75(0.48)	81.86(0.43)	85.92(0.38)	90.48(0.30)	95.16(0.12)	**98.11**(0.15)
3	50	82.05(0.52)	89.40(0.47)	87.71(0.44)	90.97(0.39)	89.28(0.42)	80.36(0.50)	**98.79**(0.13)	98.34(0.10)	97.83(0.20)
4	50	99.58(0.05)	99.94(0.02)	99.16(0.06)	99.16(0.06)	99.58(0.04)	99.91(0.02)	99.99(0.01)	**100.0**(0.01)	99.99(0.01)
5	50	95.65(0.25)	97.10(0.20)	97.52(0.18)	92.96(0.30)	96.07(0.22)	97.72(0.17)	89.37(0.36)	96.2(0.04)	**99.89**(0.05)
6	50	97.12(0.20)	98.08(0.15)	98.63(0.13)	98.22(0.14)	96.71(0.19)	97.53(0.16)	98.59(0.13)	98.29(0.06)	**99.32**(0.08)
7	12	99.87(0.02)	99.95(0.01)	99.96(0.01)	99.99(0.01)	99.92(0.02)	99.91(0.02)	99.94(0.01)	**100.0**(0.01)	99.93(0.01)
8	50	99.79(0.04)	99.96(0.02)	99.94(0.02)	99.79(0.04)	99.79(0.04)	99.98(0.01)	**99.99**(0.01)	99.54(0.01)	**99.99**(0.01)
9	12	95.00(0.45)	99.91(0.02)	99.95(0.02)	99.94(0.02)	99.93(0.02)	90.00(0.50)	99.99(0.01)	**100.0**(0.01)	99.98(0.01)
10	50	88.99(0.32)	88.48(0.34)	83.33(0.45)	85.60(0.42)	89.40(0.30)	87.35(0.38)	87.14(0.40)	**97.6**(0.18)	94.24(0.22)
11	50	72.34(1.05)	79.10(0.88)	77.88(0.91)	81.63(0.85)	86.31(0.76)	88.15(0.72)	92.71(0.55)	**94.17**(0.42)	93.44(0.52)
12	50	87.69(0.41)	87.69(0.41)	87.52(0.42)	86.68(0.44)	86.51(0.45)	91.74(0.33)	96.80(0.18)	95.04(0.06)	**99.49**(0.08)
13	50	99.92(0.01)	99.96(0.01)	99.95(0.01)	98.05(0.12)	99.91(0.01)	99.90(0.02)	99.51(0.04)	98.77(0.01)	**99.97**(0.01)
14	50	94.23(0.28)	96.36(0.22)	97.70(0.18)	96.76(0.20)	94.47(0.26)	97.08(0.17)	99.68(0.08)	**99.84**(0.06)	97.31(0.16)
15	50	98.96(0.10)	99.48(0.06)	99.22(0.08)	98.19(0.14)	99.49(0.05)	99.95(0.02)	**99.98**(0.01)	96.36(0.01)	99.74(0.04)
16	50	95.70(0.35)	99.97(0.02)	99.99(0.01)	98.92(0.12)	99.91(0.03)	95.70(0.38)	97.85(0.22)	**100.0**(0.01)	99.99(0.01)
OA(%)		87.12(0.30)	87.86(0.28)	88.55(0.26)	89.04(0.24)	90.40(0.22)	91.23(0.20)	94.14(0.18)	96.71(0.10)	**97.06**(0.12)
AA(%)		93.38(0.25)	94.31(0.22)	94.38(0.21)	94.23(0.21)	94.97(0.18)	94.47(0.20)	96.41(0.15)	98.08(0.06)	**98.73**(0.08)
κ(%)		85.50(0.32)	86.26(0.30)	87.03(0.27)	87.60(0.26)	89.10(0.23)	90.04(0.21)	93.46(0.16)	96.23(0.08)	**96.66**(0.10)
PreTime		6.0s	3.1s	5.9s	7.1s	8.0s	25.0s	4.7s	10.99s	6.6s

dataset, MSST improves AA by 0.08%, and κ by 0.28%. Furthermore, from the perspective of network generalization, MSST exhibits stable classification performance across all five datasets. In contrast, other networks show varied

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TABLE V
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE HOU13 DATASET

Class No.	Train Samples	3DCNN	SAE-CNN	IRTS-CNN	ViT	SpectralFormer	HiT	SSFTT	GraphGST	MSST
1	198	96.77(0.15)	97.25(0.12)	83.76(0.32)	96.87(0.14)	96.30(0.18)	99.15(0.05)	96.49(0.16)	**99.34**(0.04)	**99.34**(0.04)
2	190	68.23(0.28)	98.34(0.05)	95.39(0.12)	98.12(0.08)	**99.15**(0.05)	97.56(0.10)	97.09(0.11)	96.62(0.13)	98.03(0.07)
3	192	99.41(0.06)	98.80(0.04)	99.41(0.06)	98.22(0.08)	99.41(0.06)	**99.80**(0.04)	98.02(0.09)	98.60(0.05)	99.21(0.05)
4	188	85.89(0.22)	**97.54**(0.08)	92.99(0.13)	96.40(0.10)	92.42(0.13)	92.14(0.14)	94.03(0.12)	96.88(0.09)	95.84(0.13)
5	186	97.44(0.10)	95.24(0.07)	95.27(0.11)	99.24(0.07)	97.34(0.06)	99.62(0.05)	**100.0**(0.02)	99.86(0.03)	98.91(0.02)
6	182	83.92(0.20)	99.04(0.02)	96.50(0.10)	99.96(0.02)	98.60(0.06)	**100.0**(0.02)	**100.0**(0.02)	**100.0**(0.02)	**100.0**(0.02)
7	196	89.65(0.16)	82.37(0.19)	89.27(0.16)	96.08(0.10)	92.63(0.14)	92.72(0.13)	95.90(0.11)	**96.36**(0.10)	95.52(0.11)
8	191	77.49(0.23)	92.50(0.14)	93.73(0.13)	96.58(0.11)	95.63(0.11)	91.55(0.15)	**100.0**(0.02)	95.25(0.11)	98.86(0.06)
9	193	77.90(0.23)	93.58(0.12)	89.80(0.15)	97.36(0.08)	94.43(0.11)	95.47(0.10)	86.69(0.17)	96.60(0.10)	**98.96**(0.05)
10	191	87.93(0.16)	98.75(0.02)	76.45(0.21)	99.71(0.03)	99.32(0.04)	95.95(0.10)	**99.81**(0.02)	97.78(0.08)	98.36(0.07)
11	181	71.35(0.22)	70.21(0.23)	85.20(0.18)	96.49(0.11)	97.53(0.08)	97.34(0.08)	98.34(0.04)	**98.39**(0.07)	98.20(0.07)
12	192	91.35(0.14)	97.79(0.08)	89.34(0.16)	97.41(0.09)	98.56(0.06)	97.69(0.07)	99.14(0.05)	**99.90**(0.03)	99.33(0.05)
13	184	95.09(0.11)	89.12(0.16)	98.25(0.07)	99.30(0.04)	**100.0**(0.02)	92.63(0.14)	99.65(0.03)	**100.0**(0.02)	**100.0**(0.02)
14	181	99.19(0.04)	99.54(0.02)	99.89(0.02)	**100.0**(0.02)	98.38(0.07)	99.86(0.02)	**100.0**(0.02)	**100.0**(0.02)	**100.0**(0.02)
15	187	98.73(0.05)	94.50(0.12)	99.79(0.02)	99.79(0.02)	99.37(0.04)	99.58(0.03)	99.85(0.02)	**100.0**(0.02)	99.58(0.03)
OA(%)		86.10(0.11)	93.43(0.08)	90.51(0.09)	97.67(0.04)	96.89(0.06)	96.27(0.07)	97.17(0.05)	97.99(0.05)	**98.25**(0.04)
AA(%)		88.02(0.10)	94.22(0.07)	92.34(0.08)	98.10(0.04)	97.40(0.05)	96.75(0.06)	97.74(0.05)	98.44(0.04)	**98.61**(0.04)
κ(%)		84.93(0.11)	92.87(0.08)	89.70(0.09)	97.47(0.04)	96.63(0.06)	95.95(0.07)	96.93(0.05)	97.82(0.05)	**98.10**(0.04)
PreTime		18.2s	8.1s	16.1s	29.9s	25.7s	23.3s	19.7s	23.7s	20.4s

classification performances on different datasets. For example, the SpectralFormer method performs better than ViT on the SA datasets but falls short on the PU dataset. Therefore, compared to other methods, MSST demonstrates the most outstanding generalization ability.

From the information provided in Tables II–VI, the 3D-CNN, SAE-CNN, and IRTS-3DCNN utilize convolutional operations to extract spatial features from HSI; they have stronger feature extraction capabilities. However, due to the fixed geometric structure of the convolutional kernels, CNN-based methods cannot effectively capture information from distant locations in the feature maps, which, to some extent, limits their potential for further improving classification performance. The ViT network was initially designed for RGB image classification. Since HSI has a higher dimensionality compared to RGB images, directly applying ViT to HSI data results in longer sequences as inputs to the transformer. These longer sequences lead to a larger number of parameters and significantly increased computational cost, making the network susceptible to overfitting and potentially impacting the classification results. SpectralFormer, as an improved method based on ViT, outperforms traditional and CNN-based methods. However, SpectralFormer fails to fully extract joint spatial–spectral features, resulting in lower classification accuracy compared to MSST. HiT does not execute feature reduction, which limits its potential for improving classification accuracy. While SSFTT simplifies the semantic representation of features with feature tokens, the tokenization process alone is insufficient to fully capture all the useful spectral and spatial details. GraphGST, as an advanced graph-based transformer method, demonstrates superior performance compared to conventional approaches. However, GraphGST fails to sufficiently exploit the inherent spectral characteristics of HSI data and effectively integrate multiscale spatial–spectral features, leading to compromised classification accuracy when compared with MSST. It is worth noting that the significant effectiveness of the MSST method can be attributed to its ability to extract spatial and spectral correlations from HSI data through different input scales. It then fuses these correlations to obtain joint spatial–spectral features, resulting in a substantial improvement in classification accuracy.

Additionally, Tables II–VI present the prediction time (PreTime) for all methods. For all datasets, the PreTime of MSST is comparable to that of CNN-based methods, while being significantly lower than most transformer-based methods, except for SSFTT. The relatively low PreTime of SSFTT is attributed to its use of PCA for dimensionality reduction of the input data prior to feature extraction and classification. However, this preprocessing step inevitably disrupts the intrinsic spatial–spectral structure of HSIs. In contrast, the proposed MSST method adopts multiscale sampling as input, preserving the original spatial–spectral feature integrity of the HSI data, resulting in higher classification accuracy.

The classification results of SAE-CNN, 3D-CNN, IRTS-3DCNN, ViT, SpectralFormer, HiT, SSFTT, GraphGST, and the MSST method are presented in Figs. 10–14. Almost all methods demonstrate effective classification of HSI. However, when compared to MSST, the classification maps generated by other methods exhibit more significant noise and misclassifications. It is worth noting that the MSST method provides clearer boundaries on all three datasets compared to other networks, allowing for a distinct differentiation of different

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TABLE VI
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE HOU18 DATASET

Class No.	Train Samples	3DCNN	SAE-CNN	IRTS-CNN	ViT	SpectralFormer	HiT	SSFTT	GraphGST	MSST
1	100	88.92(0.21)	60.03(0.34)	91.03(0.15)	77.57(0.28)	83.15(0.22)	89.09(0.18)	79.83(0.26)	**91.33**(0.14)	89.81(0.17)
2	500	92.50(0.15)	87.46(0.19)	93.06(0.12)	94.42(0.11)	94.80(0.10)	95.60(0.08)	94.06(0.09)	92.46(0.11)	**95.60**(0.08)
3	100	99.74(0.05)	99.52(0.06)	99.34(0.06)	99.60(0.04)	99.67(0.04)	99.45(0.05)	99.82(0.03)	99.76(0.04)	**100.0**(0.02)
4	500	98.05(0.08)	96.82(0.10)	97.28(0.09)	98.50(0.07)	97.96(0.08)	**98.90**(0.07)	97.53(0.09)	94.83(0.12)	97.73(0.08)
5	100	77.26(0.22)	50.55(0.33)	78.23(0.21)	82.07(0.19)	85.54(0.17)	82.90(0.20)	89.96(0.15)	**91.00**(0.13)	89.66(0.14)
6	100	91.43(0.17)	91.70(0.14)	91.69(0.16)	98.08(0.08)	98.28(0.07)	96.99(0.09)	97.12(0.09)	**100.0**(0.02)	98.94(0.06)
7	50	92.20(0.13)	93.52(0.12)	98.68(0.06)	**100.0**(0.02)	**100.0**(0.02)	99.91(0.03)	99.81(0.03)	**100.0**(0.02)	**100.0**(0.02)
8	500	82.54(0.20)	85.01(0.18)	84.56(0.19)	88.63(0.15)	89.49(0.14)	86.84(0.16)	87.55(0.16)	**92.97**(0.10)	89.17(0.14)
9	500	81.44(0.19)	67.11(0.25)	77.51(0.22)	86.23(0.16)	85.98(0.17)	84.68(0.18)	85.09(0.17)	**93.81**(0.09)	86.56(0.15)
10	500	51.17(0.28)	50.21(0.30)	59.72(0.26)	69.53(0.21)	**69.99**(0.21)	67.28(0.23)	65.11(0.24)	69.20(0.21)	69.43(0.21)
11	500	67.60(0.23)	64.61(0.24)	69.44(0.22)	71.65(0.19)	71.51(0.20)	73.73(0.18)	**75.69**(0.17)	72.97(0.19)	73.96(0.18)
12	100	36.79(0.32)	41.57(0.30)	31.54(0.34)	52.75(0.25)	56.31(0.23)	35.76(0.32)	50.49(0.26)	56.00(0.24)	**61.02**(0.20)
13	500	75.55(0.21)	67.98(0.24)	66.54(0.25)	79.58(0.18)	77.91(0.19)	68.69(0.23)	79.18(0.19)	**80.51**(0.18)	77.29(0.20)
14	100	80.94(0.18)	69.05(0.25)	73.78(0.22)	82.48(0.17)	83.98(0.16)	85.12(0.15)	80.59(0.18)	**85.79**(0.14)	84.04(0.15)
15	100	97.39(0.08)	94.41(0.10)	93.84(0.11)	97.86(0.07)	98.01(0.07)	97.26(0.08)	97.31(0.08)	**98.10**(0.06)	97.77(0.07)
16	500	94.58(0.09)	93.35(0.10)	92.27(0.11)	96.82(0.08)	97.15(0.07)	97.75(0.07)	97.24(0.07)	97.82(0.07)	**97.96**(0.06)
17	50	99.83(0.03)	99.66(0.03)	98.30(0.05)	**100.0**(0.02)	**100.0**(0.02)	**100.0**(0.02)	99.49(0.03)	**100.0**(0.02)	**100.0**(0.02)
18	100	78.28(0.19)	71.22(0.22)	72.75(0.20)	85.98(0.14)	90.46(0.11)	85.45(0.14)	**92.51**(0.09)	91.60(0.10)	86.21(0.14)
19	100	79.51(0.20)	80.33(0.19)	79.44(0.20)	92.45(0.12)	90.79(0.13)	94.18(0.09)	91.52(0.11)	**98.13**(0.06)	95.96(0.08)
20	100	90.58(0.14)	83.02(0.17)	87.69(0.15)	97.35(0.08)	97.54(0.08)	97.06(0.08)	96.11(0.09)	**100.0**(0.02)	98.73(0.06)
OA(%)		79.13(0.12)	70.33(0.16)	77.32(0.13)	84.54(0.09)	84.62(0.09)	82.97(0.10)	83.21(0.10)	**85.97**(0.08)	85.18(0.08)
AA(%)		82.82(0.11)	77.36(0.13)	81.83(0.12)	87.02(0.09)	88.43(0.08)	86.83(0.09)	87.50(0.09)	89.41(0.07)	**89.49**(0.07)
κ(%)		73.75(0.13)	63.94(0.17)	71.74(0.14)	80.42(0.10)	80.55(0.09)	78.53(0.11)	79.28(0.10)	80.85(0.09)	**81.23**(0.09)
PreTime		67.58s	40.8s	52.42s	98.05s	53.68s	50.87s	42.21s	81.12s	70.20s

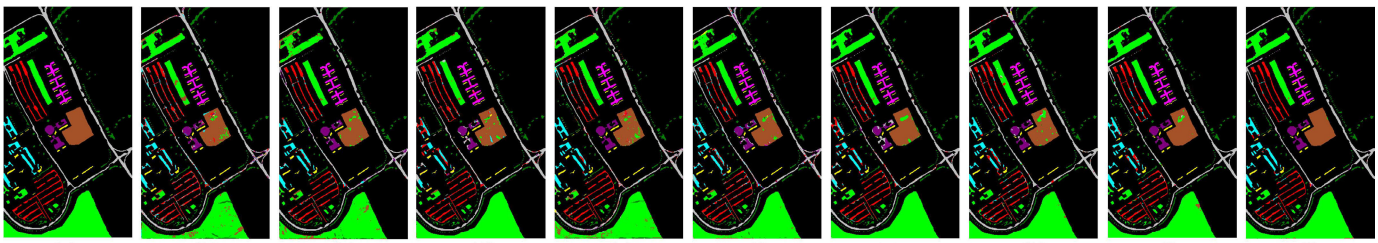


Fig. 10. Classification result of different methods using the PU dataset without BKG. (a) Ground truth. (b) 3D-CNN. (c) SAE-CNN. (d) IRTS-3DCNN. (e) ViT. (f) SpectralFormer. (g) HiT. (h) SSFTT. (i) GraphGST. (j) MSST.

types of land cover targets and achieving results closer to the ground truth. This superior performance is attributed to the excellent extraction capability of SSM and SpecFormer branch, which enhances the extraction of spectral structural features.

In HSI classification tasks, complicated BKG regions often pose significant challenges. To verify the performance of MSST under complicated BKG conditions, we conducted experiments on the above four datasets. For evaluation, we adopted the metrics proposed in IRTS-CNN [30]: precision rate (PR), overall PR (P_{opr}), and average PR (P_{apr}). PR measures the classification precision for each class, defined as the ratio of correctly classified samples to the total predicted samples for that class. P_{opr}, as the overall PR, reflects the weighted precision across all classes. P_{apr} is the arithmetic mean of the PRs for all classes.

The classification results of various methods on the four datasets under BKG conditions are shown in Tables VII–IX. When considering BKG, the theoretical maximum P_{opr} achievable on the PU, SA, and IP datasets is 20.62%, 48.72%, and 48.75%, respectively. Specifically, the PU dataset has prominent BKG issues, which makes it an ideal benchmark for evaluating the adaptability and robustness of classification models under such conditions. On the PU dataset, MSST achieves the highest PR for Class No.2, which contains the largest number of samples. These results strongly validate

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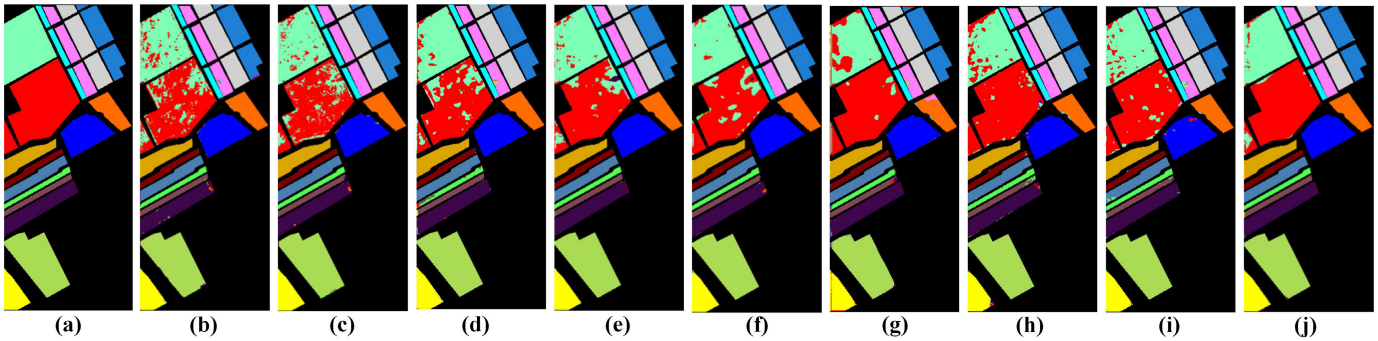


Fig. 11. Classification result of different methods using the SA dataset without BKG. (a) Ground truth. (b) 3D-CNN. (c) SAE-CNN. (d) IRTS-3DCNN. (e) ViT. (f) SpectralFormer. (g) HiT. (h) SSFTT. (i) GraphGST. (j) MSST.

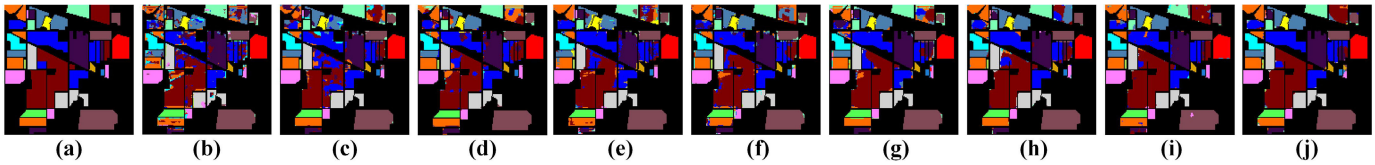


Fig. 12. Classification result of different methods using the IP dataset without BKG. (a) Ground truth. (b) 3D-CNN. (c) SAE-CNN. (d) IRTS-3DCNN. (e) ViT. (f) SpectralFormer. (g) HiT. (h) SSFTT. (i) GraphGST. (j) MSST.

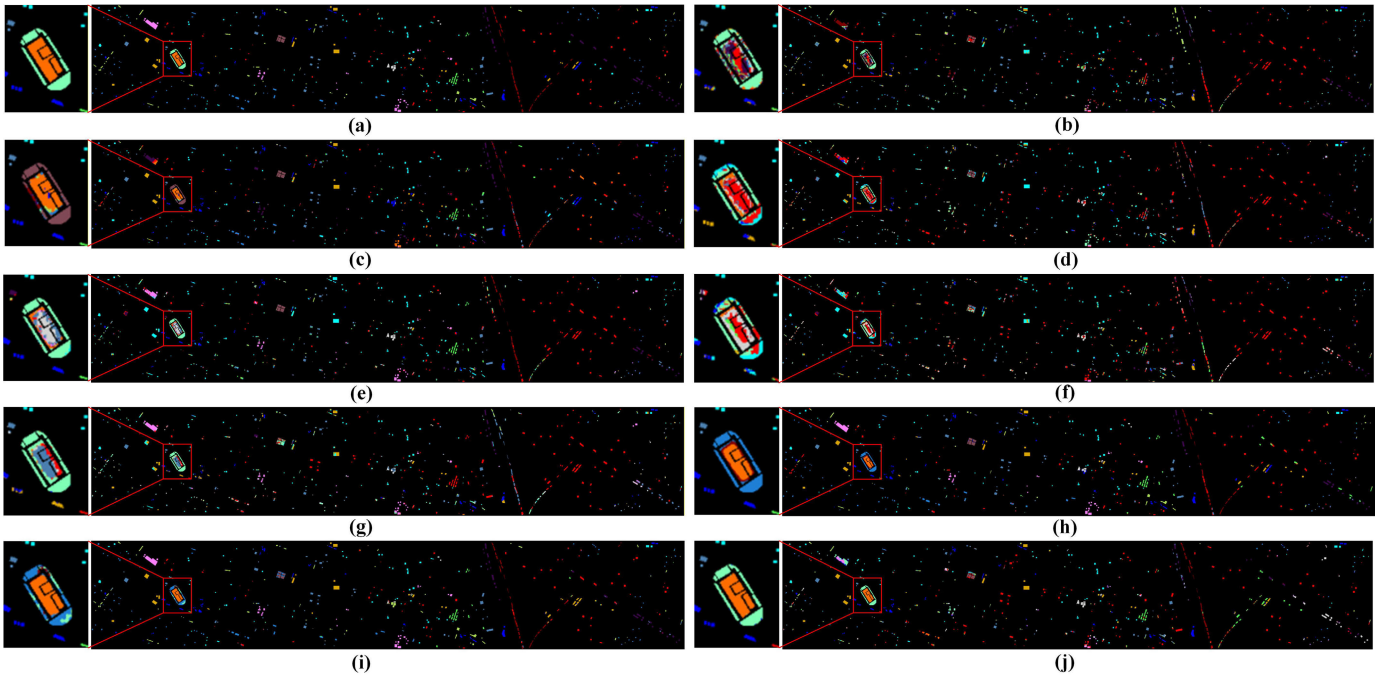


Fig. 13. Classification result of different methods using the Hou13 dataset without BKG. (a) Ground truth. (b) 3D-CNN. (c) SAE-CNN. (d) IRTS-3DCNN. (e) ViT. (f) SpectralFormer. (g) HiT. (h) SSFTT. (i) GraphGST. (j) MSST.

the effectiveness of the MSST architecture in handling BKG issues while preserving spatial–spectral features. To provide a visual inspection of the test methods, Figs. 15–17 present their classification maps of different methods on the four datasets under BKG conditions. It can be observed that MSST produces more coherent and precise class boundaries, particularly in complex regions where other methods tend to generate noisy or fragmented predictions. For example, in the PU dataset, the separation between spectrally similar classes, such as Class 2 Meadows and Class 6 Bare Soil, aligns more closely with the visual distribution in the RGB reference image. This visual consistency, observed across all four datasets, highlights the strong generalization ability of the MSST method under realistic classification scenarios involving BKG issues.

In combination with prior research [27], we discovered a striking similarity in the spectral characteristics of class 8 “Grapes Untrained” and class 15 “Vineyard Untrained” in the SA dataset, as depicted in Fig. 18. In the figure, the red

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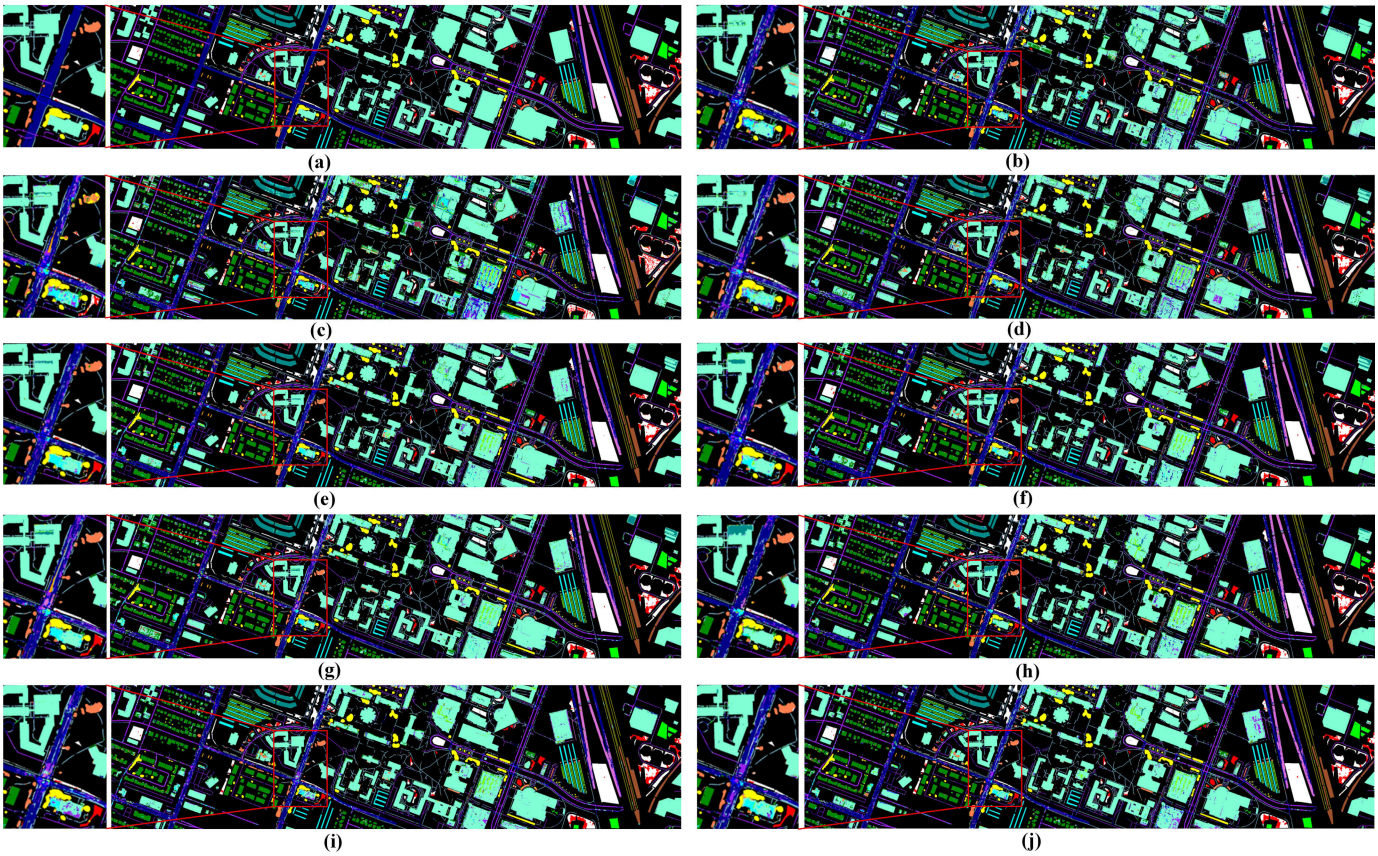


Fig. 14. Classification result of different methods using the Hou18 dataset without BKG. (a) Ground truth. (b) 3D-CNN. (c) SAE-CNN. (d) IRTS-3DCNN. (e) ViT. (f) SpectralFormer. (g) HiT. (h) SSFTT. (i) GraphGST. (j) MSST.

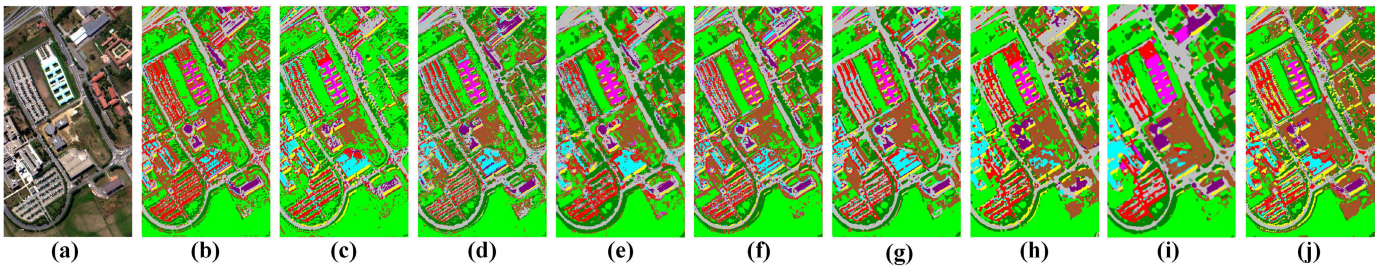


Fig. 15. Classification result of different methods using the PU dataset with BKG. (a) RGB image. (b) 3D-CNN. (c) SAE-CNN. (d) IRTS-3DCNN. (e) ViT. (f) SpectralFormer. (g) HiT. (h) SSFTT. (i) GraphGST. (j) MSST.

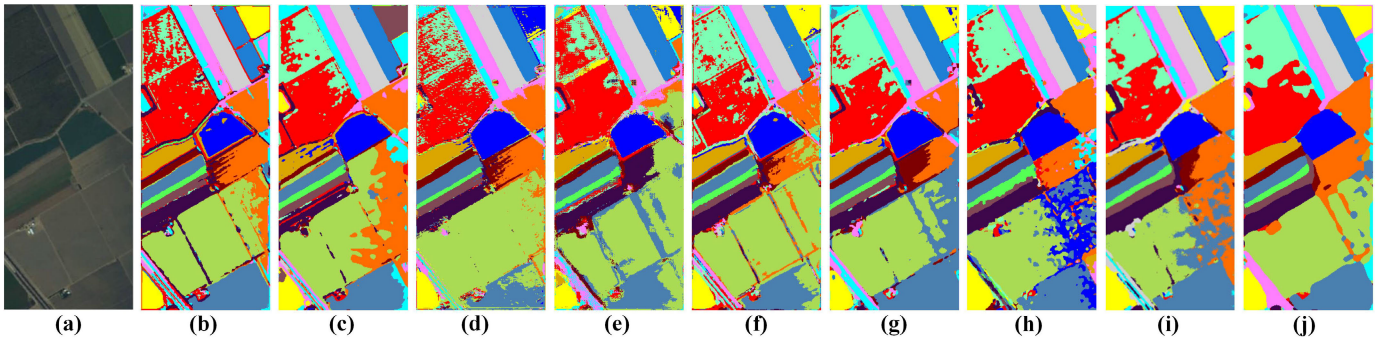


Fig. 16. Classification result of different methods using the SA dataset with BKG. (a) RGB image. (b) 3D-CNN. (c) SAE-CNN. (d) IRTS-3DCNN. (e) ViT. (f) SpectralFormer. (g) HiT. (h) SSFTT. (i) GraphGST. (j) MSST.

and blue curves, respectively, depict the spectral signatures of a single pixel drawn from classes 8 and 15, with the horizontal axis denoting the spectral bands and the vertical axis indicating the corresponding reflectance values. This

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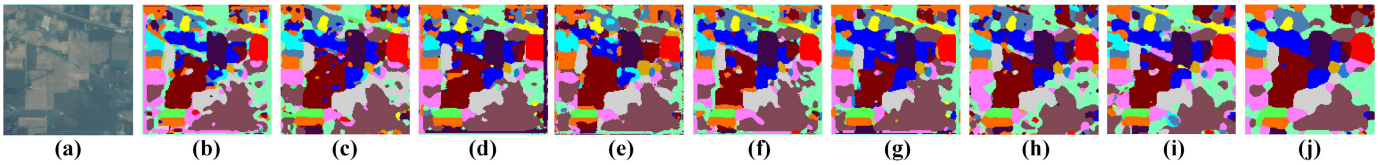


Fig. 17. Classification result of different methods using the IP dataset with BKG. (a) RGB image. (b) 3D-CNN. (c) SAE-CNN. (d) IRTS-3DCNN. (e) ViT. (f) SpectralFormer. (g) HiT. (h) SSFTT. (i) GraphGST. (j) MSST.

TABLE VII
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE PU DATASET WITH BKG

Class No.	Train Samples	3D-CNN	SAE-CNN	IRTS-CNN	ViT	Spectral-Former	HiT	SSFTT	GraphGST	MSST
1	66	**21.04**(0.76)	18.75(0.81)	14.09(0.77)	15.87(0.63)	19.26(0.69)	18.07(0.75)	18.62(0.68)	15.48(0.53)	20.91(0.32)
2	186	23.82(0.91)	21.04(0.85)	36.90(0.84)	34.18(0.76)	32.83(0.79)	32.08(0.31)	31.41(0.74)	29.30(0.63)	**46.66**(0.29)
3	20	**27.02**(0.89)	14.21(0.33)	11.12(0.84)	12.88(0.90)	9.71(0.87)	13.14(0.82)	20.80(0.85)	26.54(0.78)	19.18(0.35)
4	30	**12.30**(0.77)	11.45(0.74)	7.94(0.20)	6.14(0.53)	9.69(0.69)	9.45(0.51)	9.12(0.40)	8.08(0.32)	8.15(0.30)
5	13	43.27(0.91)	49.34(0.86)	44.49(0.90)	30.96(0.77)	20.33(0.80)	37.94(0.78)	42.66(1.04)	20.55(0.96)	**55.54**(0.26)
6	50	7.58(0.45)	**19.07**(0.92)	9.82(0.77)	17.07(0.99)	9.39(0.81)	11.57(0.72)	12.26(0.38)	18.27(0.41)	9.75(0.31)
7	13	24.21(0.84)	14.56(0.86)	12.16(0.75)	18.02(0.10)	17.27(0.77)	14.48(0.24)	15.87(0.72)	16.00(0.30)	**29.27**(0.33)
8	36	15.24(0.88)	16.51(0.83)	19.49(0.85)	10.30(0.78)	17.30(0.81)	18.02(0.29)	18.37(0.76)	17.95(0.49)	**19.98**(0.90)
9	9	**29.60**(0.07)	10.58(0.89)	22.57(1.14)	21.38(0.80)	10.87(0.82)	12.88(0.78)	7.86(0.15)	14.85(0.63)	15.65(0.28)
P_{opr}		18.02(0.38)	18.35(0.29)	18.34(0.45)	18.42(0.50)	18.41(0.73)	19.29(0.68)	19.51(0.56)	19.57(0.57)	**19.81**(0.24)
P_{apr}		22.68(0.79)	19.50(0.76)	19.84(0.77)	18.53(0.32)	16.29(0.71)	18.63(0.69)	19.67(0.55)	18.56(0.51)	**24.62**(0.22)

TABLE VIII
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE SA DATASET WITH BKG

Class No.	Train Samples	3D-CNN	SAE-CNN	IRTS-CNN	ViT	Spectral-Former	HiT	SSFTT	GraphGST	MSST
1	20	69.74(4.12)	60.92(5.76)	59.14(5.69)	73.48(3.89)	**84.78**(0.65)	71.93(3.76)	67.43(4.28)	80.96(1.17)	65.89(4.92)
2	37	**93.69**(0.32)	79.36(1.74)	51.29(7.85)	76.99(1.69)	79.50(1.84)	83.14(1.43)	29.95(8.78)	69.37(2.23)	76.00(1.98)
3	20	16.94(9.75)	12.38(11.95)	20.57(9.83)	**29.17**(7.39)	20.16(9.51)	28.64(7.91)	20.84(9.45)	14.27(8.91)	16.84(7.24)
4	14	24.58(9.87)	21.46(10.95)	40.83(7.01)	33.45(8.23)	**45.70**(6.53)	27.16(8.76)	39.78(7.41)	31.88(7.92)	28.53(8.67)
5	27	47.86(6.53)	46.53(6.84)	38.66(7.76)	38.24(7.95)	41.79(7.12)	35.88(8.11)	45.87(6.63)	**57.44**(5.98)	39.26(7.44)
6	40	74.04(3.08)	77.83(2.57)	**84.36**(0.94)	69.19(4.57)	63.40(5.12)	84.07(1.02)	62.58(5.34)	48.35(6.10)	81.29(1.48)
7	36	**88.88**(0.55)	84.00(1.02)	79.31(1.68)	70.79(3.92)	79.16(1.86)	80.84(1.53)	79.39(1.74)	88.71(0.60)	78.63(1.93)
8	113	53.90(7.05)	67.07(5.01)	67.74(4.85)	58.14(6.78)	67.47(4.94)	77.58(2.45)	74.91(2.89)	**84.51**(2.10)	77.65(2.39)
9	33	27.99(9.25)	29.25(8.86)	20.62(10.58)	18.70(11.72)	21.74(9.89)	31.94(8.32)	37.62(7.41)	**43.46**(6.85)	26.28(6.78)
10	11	38.12(8.94)	55.99(5.47)	67.75(4.76)	41.39(8.11)	68.70(4.39)	61.03(5.42)	56.22(5.73)	51.89(6.10)	**80.05**(1.46)
11	19	32.78(8.46)	29.48(9.03)	31.85(8.39)	24.56(9.48)	32.88(8.24)	22.74(9.62)	**65.07**(4.61)	32.62(8.10)	49.90(6.17)
12	9	20.02(9.65)	21.84(9.34)	24.03(8.92)	14.74(11.85)	22.82(9.21)	10.11(10.76)	14.55(7.28)	11.23(9.00)	**53.00**(5.29)
13	27	50.06(6.54)	63.85(4.53)	**85.28**(0.88)	56.88(5.15)	62.86(4.79)	70.31(3.48)	44.65(7.08)	67.28(4.03)	71.19(3.18)
14	11	53.41(5.76)	23.71(10.21)	56.78(5.26)	**65.24**(4.32)	63.70(4.56)	60.98(4.89)	63.99(4.67)	53.96(5.15)	53.73(5.94)
15	73	74.44(3.17)	74.19(3.26)	60.70(5.82)	67.50(4.63)	55.53(6.78)	66.63(4.51)	74.93(3.05)	**79.41**(2.80)	75.97(2.88)
16	18	41.80(7.52)	**67.45**(4.76)	66.56(4.81)	33.72(8.62)	37.61(8.15)	39.04(7.81)	44.86(7.24)	29.52(8.50)	39.75(1.18)
P_{opr}		42.89(0.57)	44.03(0.64)	43.81(0.79)	40.11(0.83)	44.66(0.71)	45.32(0.68)	46.40(0.73)	45.53(0.70)	**46.81**(0.59)
P_{apr}		50.52(0.53)	50.96(0.77)	53.47(0.48)	48.26(0.85)	52.99(0.66)	53.25(0.72)	51.41(0.58)	52.80(0.65)	**54.86**(0.61)

resemblance presents a considerable challenge for feature extraction and classification. To validate the feature extraction capability of our method, we utilize t-distributed stochastic neighbor embedding (t-SNE) to visualize and analyze the feature mappings obtained by different methods for class 8 and class 15 data.

As illustrated in Fig. 19. We have compared MSST with seven other competitive methods (3D-CNN, SAE-CNN,

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TABLE IX
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS ON THE IP DATASET WITH BKG

Class No.	Train Samples	3D-CNN	SAE-CNN	IRTS-CNN	ViT	Spectral-Former	HiT	SSFTT	GraphGST	MSST
1	25	45.53(2.24)	28.05(3.53)	46.00(2.13)	9.47(1.42)	**47.87**(2.03)	23.23(3.84)	20.18(4.15)	31.08(3.12)	30.46(3.12)
2	50	61.19(1.83)	61.43(1.82)	62.20(1.71)	51.76(2.13)	64.91(1.52)	50.05(2.15)	66.50(1.41)	**73.61**(1.21)	68.47(1.32)
3	50	44.04(2.13)	48.68(2.42)	45.80(2.83)	28.05(3.62)	40.25(2.73)	42.79(2.33)	61.84(1.91)	**72.18**(1.92)	60.38(2.02)
4	50	29.44(3.62)	36.69(2.33)	**71.63**(1.41)	26.33(3.83)	41.09(2.63)	67.39(1.71)	54.76(2.02)	45.14(2.42)	59.80(2.12)
5	50	21.53(3.84)	**36.20**(2.03)	24.81(3.53)	30.39(3.62)	27.10(3.53)	30.91(3.52)	24.25(3.73)	20.73(3.12)	23.47(3.94)
6	50	38.43(2.73)	27.49(3.53)	49.35(2.83)	37.35(2.33)	48.23(2.03)	**66.89**(1.61)	42.38(2.43)	43.22(2.33)	46.88(2.13)
7	12	62.79(2.02)	36.49(2.33)	49.12(2.03)	21.54(3.84)	**68.42**(1.42)	16.97(3.87)	29.17(3.62)	31.08(3.12)	21.71(3.94)
8	50	76.52(1.33)	53.89(2.42)	**84.30**(1.63)	78.23(1.43)	65.75(1.91)	77.91(1.43)	59.45(2.52)	51.18(2.83)	62.24(2.12)
9	12	22.35(3.84)	38.46(2.23)	**43.48**(2.43)	25.00(3.53)	18.02(3.87)	21.74(3.94)	15.27(4.42)	12.27(3.87)	32.26(3.52)
10	50	66.70(1.41)	60.04(2.02)	38.49(2.13)	47.95(2.83)	**70.87**(1.32)	65.96(1.61)	55.85(2.82)	65.62(1.91)	63.37(1.91)
11	50	77.68(1.33)	71.94(1.32)	80.14(1.93)	54.16(2.42)	78.85(1.23)	**89.07**(0.83)	77.05(1.33)	88.88(1.02)	76.29(1.43)
12	50	48.76(2.83)	53.45(2.42)	42.41(2.73)	35.97(2.33)	46.88(2.13)	**73.68**(1.41)	40.49(2.73)	42.93(2.42)	58.06(2.52)
13	50	50.49(2.83)	36.22(2.33)	49.40(2.83)	67.33(1.71)	45.35(2.43)	50.37(2.83)	60.18(2.12)	**67.77**(1.61)	48.34(2.13)
14	50	31.40(3.52)	30.11(3.62)	30.70(3.52)	25.93(3.53)	30.40(3.52)	25.16(3.73)	**40.57**(2.73)	35.27(2.83)	36.08(2.33)
15	50	10.93(3.42)	19.81(3.75)	16.61(3.32)	15.05(3.32)	12.02(3.42)	15.68(3.32)	11.94(3.42)	14.62(3.12)	**20.86**(3.52)
16	50	22.91(3.84)	30.20(3.62)	15.40(3.32)	40.81(2.73)	19.13(3.87)	15.76(3.32)	35.91(2.13)	12.64(3.87)	**52.33**(2.42)
P_{opr}		40.85(1.23)	43.22(1.32)	43.15(1.22)	36.77(2.04)	42.67(1.33)	43.98(1.14)	45.17(1.22)	**46.13**(1.15)	45.95(0.85)
P_{apr}		45.92(1.14)	41.82(1.32)	46.86(1.05)	37.83(1.82)	45.32(1.24)	45.85(1.14)	43.49(1.13)	44.26(1.20)	**46.94**(0.94)

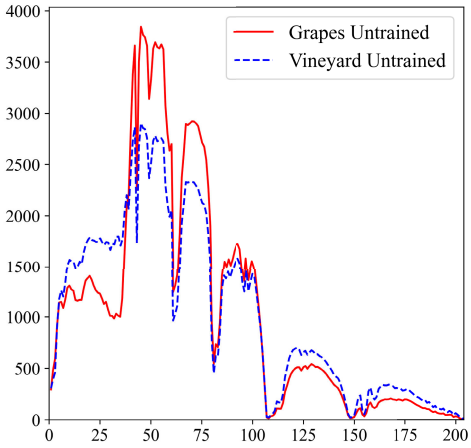


Fig. 18. Spectral signatures of the eighth and fifteenth categories of SA dataset.

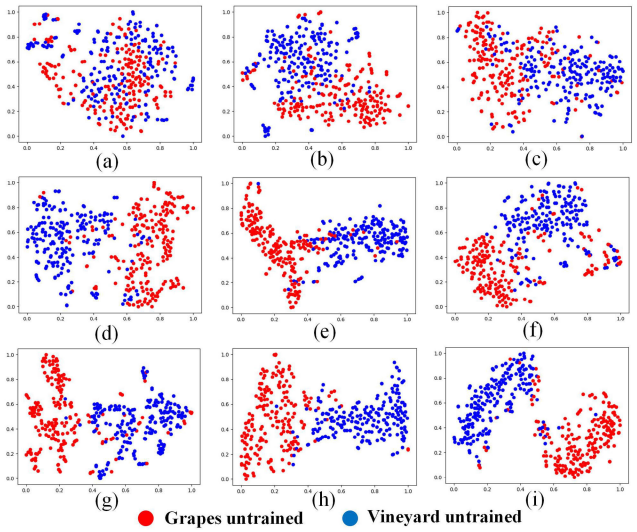


Fig. 19. t-SNE visualization after feature extraction with different methods. (a) 3D-CNN. (b) SAE-CNN. (c) IRTS-3DCNN. (d) ViT. (e) SpectralFormer. (f) HiT. (g) SSFTT. (h) GraphGST. (i) MSST.

TABLE X
CLUSTERING CONDITION AFTER FEATURE EXTRACTION WITH DIFFERENT METHODS

Methods	ViT	SF	HiT	SSFTT	GraphGST	MSST
Intra-class Variance	0.093	0.078	0.087	0.083	0.068	**0.059**
Inter-class Variance	0.158	0.212	0.155	0.197	0.234	**0.249**

IRTS-3DCNN, ViT, SpectralFormer, HiT, SSFTT, and GraphGST). It is clear that 3D-CNN, SAE-CNN, and IRTS-3DCNN exhibit limited discriminative ability for these two classes of features, while the feature maps generated by the transformer-based methods are linearly separable. However, the clustering capability of the transformer-based methods still exhibits some variations. To further investigate this, we precisely calculated the intraclass variance and interclass variance for these two classes of features, as shown in Table X. The results indicate that MSST exhibits the lowest intra-class variance and the highest interclass variance. Moreover, the SSM and SpecFormer modules enhance the separability between features with similar spectral structure, making it easier for the MSST to extract joint spatial–spectral features and

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improving classification performance by avoiding interference from similar classes.

IV. Conclusion

In this article, we propose a novel transformer-based MSST network designed for efficient HSI classification. The MSST network is an end-to-end architecture that takes spectral samples as input at two different scales. It leverages both SpecSAM and SpatSAM mechanisms to extract spatial and spectral structural features from the samples. These features are then fused to enhance the overall classification performance. Specifically, on the spectral branch, the SSM module is used to perform overlapped sampling on the spectral data, and the SpecSAM is employed to find spectral structural correlations among short spectral sequences. On the spatial branch, the CDRM module is used for dimension reduction and strengthening the representation of spatial features. Subsequently, the SpatSAM is used to extract remote spatial–spectral structural features from the HSI data. Finally, the features from both branches are effectively combined and utilized for accurate classification.

To evaluate the effectiveness of our proposed method, extensive experiments are conducted on the PU, SA, IP, Hou13, and Hou18 datasets. In comparison with other state-of-the-art methods, MSST achieves the best classification results and demonstrates strong generalization ability. In future work, we will focus on streamlining the network architecture while ensuring the preservation of classification accuracy. This will allow us to create more efficient and practical models for HSI classification tasks.

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